AI VIETNAM
All-in-One Course
(TA Session)

Streamlit and Gradio

Extra Class: MLOps

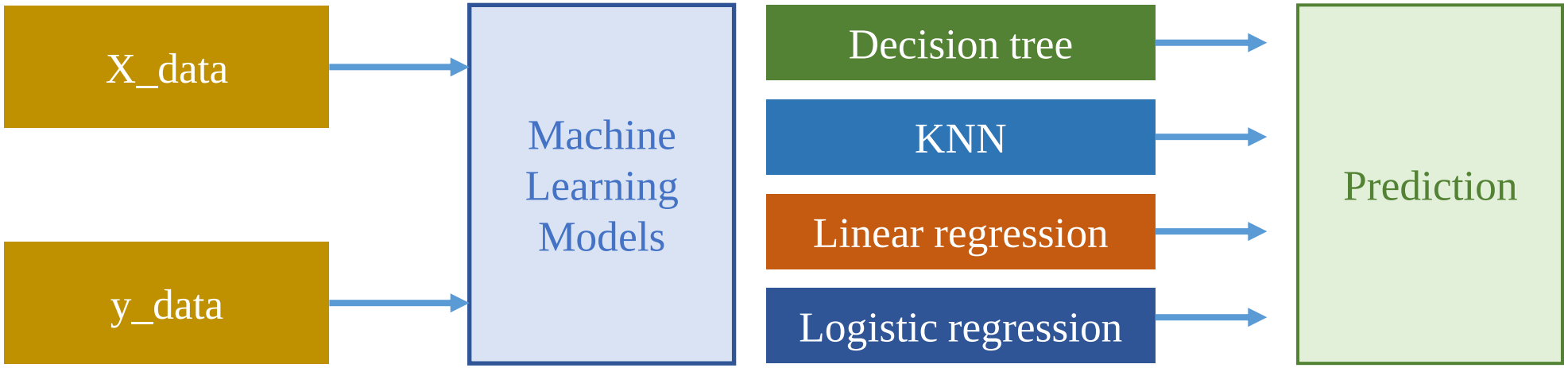


Nguyen-Thuan Duong – TA

Year 2024

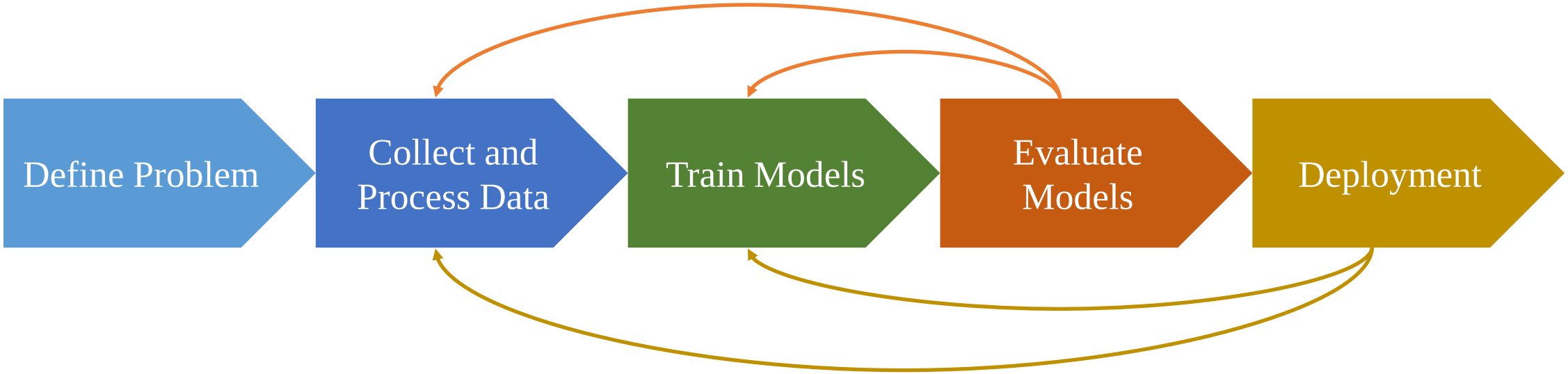
Machine Learning

❖ Machine Learning Models



Machine Learning

❖ Machine Learning Life Cycle



Machine Learning

❖ Machine Learning Operation (MLOps)

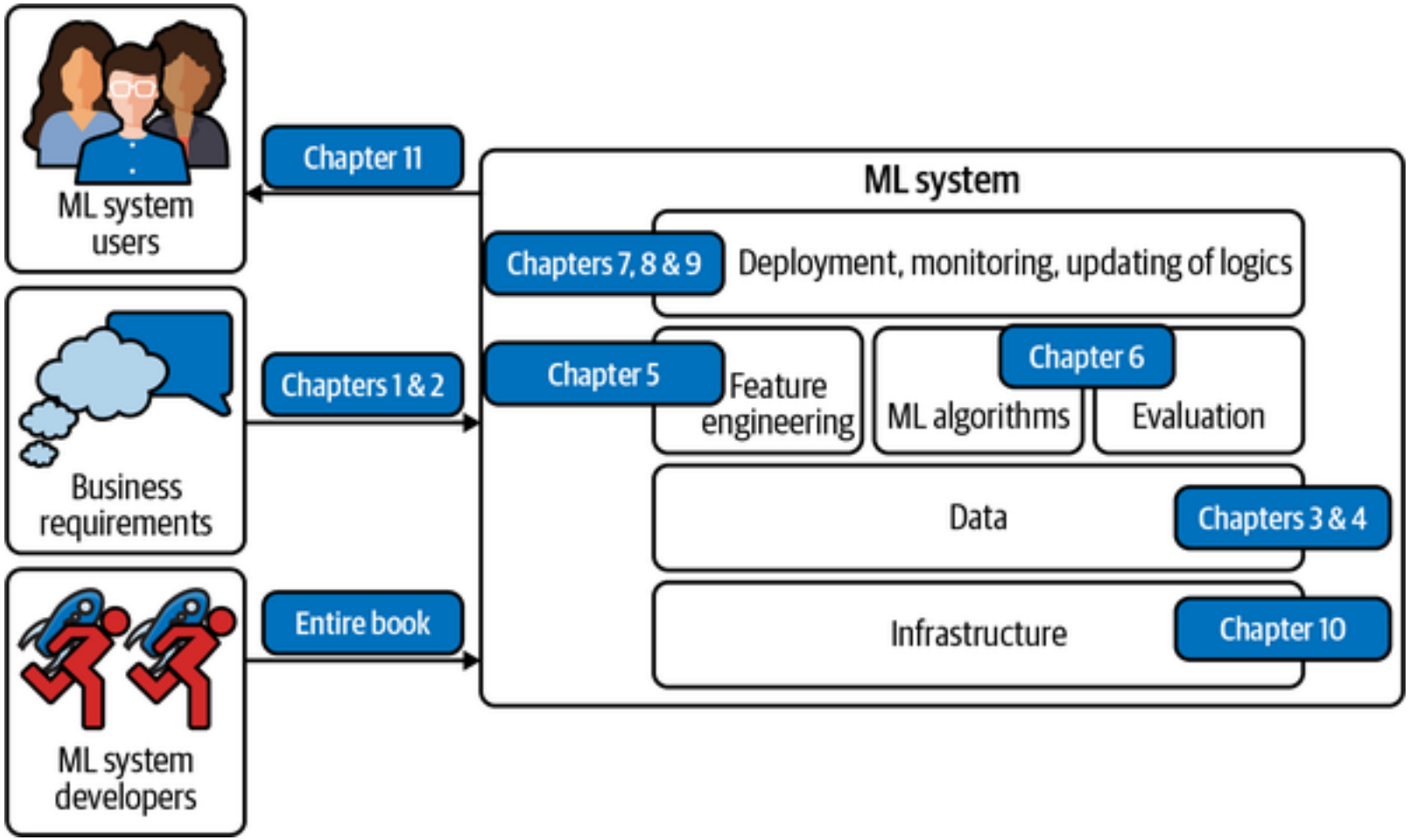
- MLOps is a set of tools and best practices for bringing ML into production.



Ref: Designing Machine Learning System – Chip Huyen

Machine Learning

❖ Machine Learning Systems



Ref: Designing Machine Learning System – Chip Huyen

Outline

- Introduction
- Web Application
- Streamlit
- Gradio
- Hands-on
- Conclusion
- Question

Introduction

Introduction

❖ Getting Started

```python
from transformers import Wav2Vec2Processor, Wav2Vec2ForCTC
import soundfile as sf
import torch

processor = Wav2Vec2Processor.from_pretrained("nguyenvulebinh/wav2vec2-base-vietnamese-250h")
model = Wav2Vec2ForCTC.from_pretrained("nguyenvulebinh/wav2vec2-base-vietnamese-250h")

# define function to read in sound file
def map_to_array(batch):
    speech, _ = sf.read(batch["file"])
    batch["speech"] = speech
    return batch

# load dummy dataset and read soundfiles
ds = map_to_array({
    "file": 'audio-test/t1_0001-00010.wav'
})

input_values = processor(ds["speech"], return_tensors="pt", padding="longest").input_values
logits = model(input_values).logits

# take argmax and decode
predicted_ids = torch.argmax(logits, dim=-1)
transcription = processor.batch_decode(predicted_ids)
```



Introduction

❖ Developer vs End-user



Here is my new model





Dev



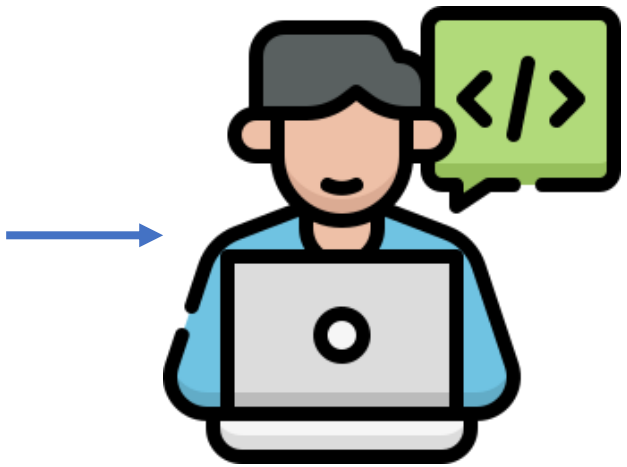
End-user

Introduction

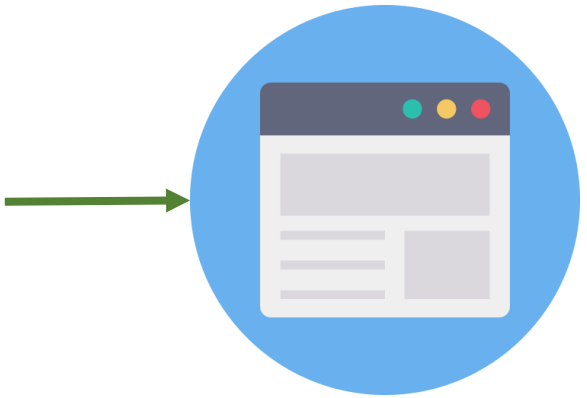
❖ Development







Developer



UI/App

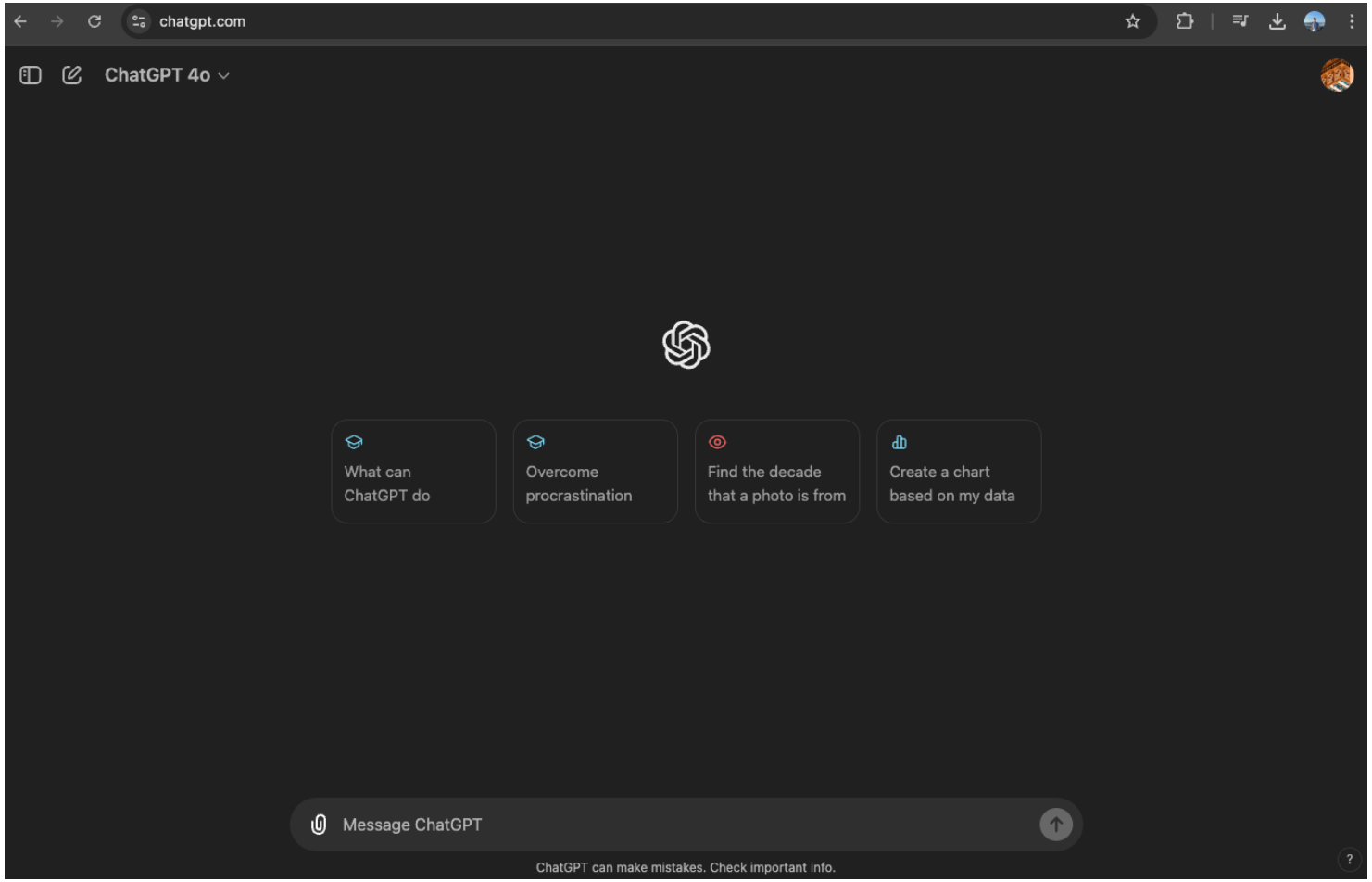


End-user

Web application

Web application

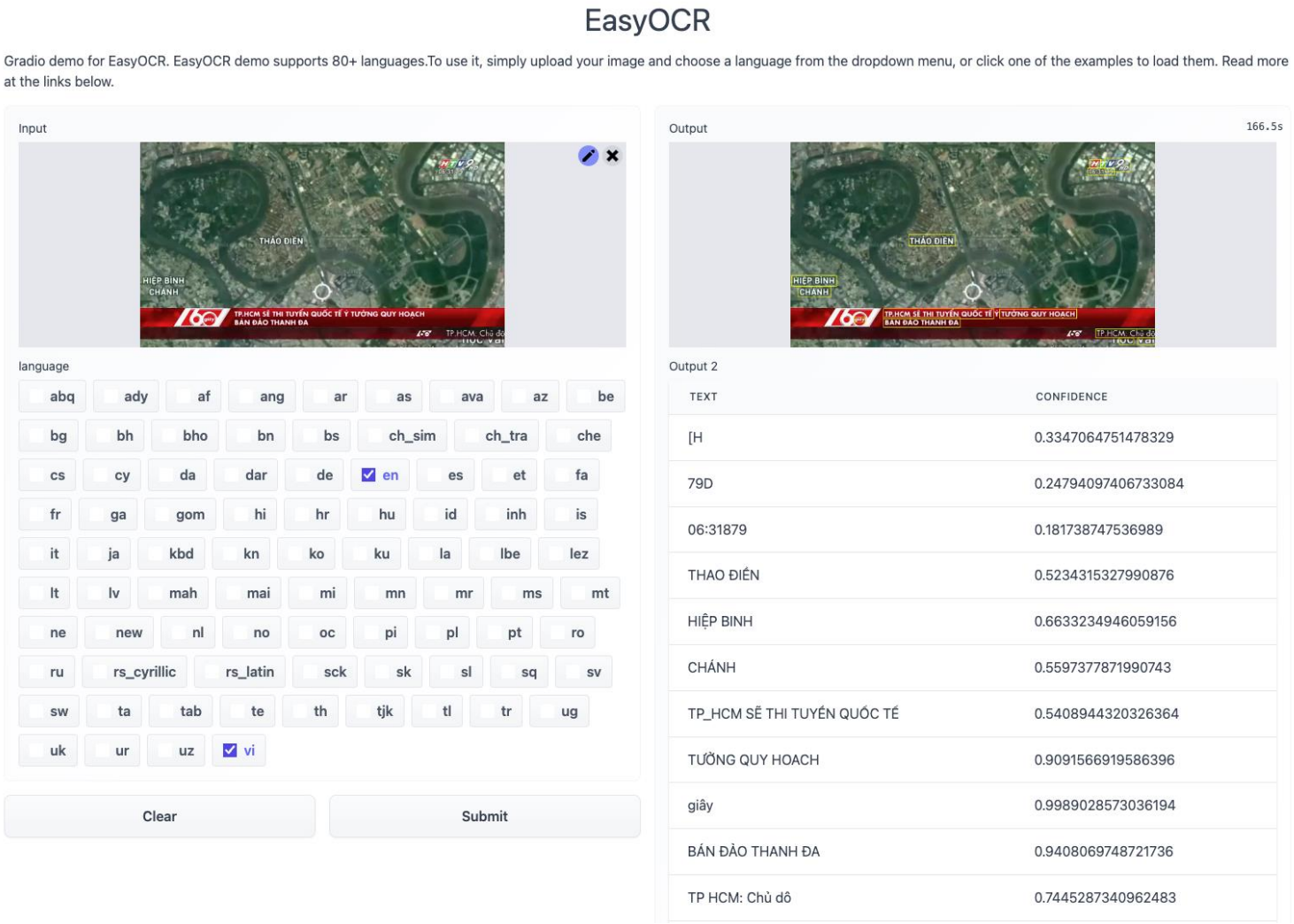
❖ ChatGPT web application



Ref: https://chatgpt.com

Introduction

❖ OCR demo



Ref: https://huggingface.co/spaces/tomofi/EasyOCR

Introduction

❖ Web application



Introduction

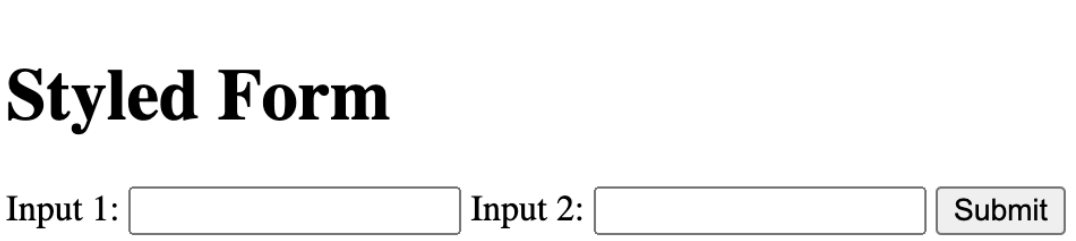
❖ HTML

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Styled Text Input Form</title>
</head>
<body>

    <div class="container">
        <h1>Styled Form</h1>
        <form action="#" method="post">
            <label for="input1">Input 1:</label>
            <input type="text" id="input1" name="input1" required>
            <label for="input2">Input 2:</label>
            <input type="text" id="input2" name="input2" required>
            <button type="submit">Submit</button>
        </form>
    </div>

</body>
</html>

```



Introduction

❖ HTML + CSS

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Styled Text Input Form</title>
    <link rel="stylesheet" href="index.css">
</head>
<body>

    <div class="container">
        <h1>Styled Form</h1>
        <form action="#" method="post">
            <label for="input1">Input 1:</label>
            <input type="text" id="input1" name="input1" required>
            <label for="input2">Input 2:</label>
            <input type="text" id="input2" name="input2" required>
            <button type="submit">Submit</button>
        </form>
    </div>

</body>
</html>

```

Introduction

❖ HTML + CSS

```css
body {
    font-family: Arial, sans-serif;
    background-color: #f4f4f9;
    margin: 0;
    padding: 0;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

h1 {
    text-align: center;
    color: #333;
}
```

```css
form {
    background-color: #fff;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
    width: 100%;
    max-width: 400px;
}

label {
    font-size: 16px;
    color: #555;
}

input[type="text"] {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
}
```

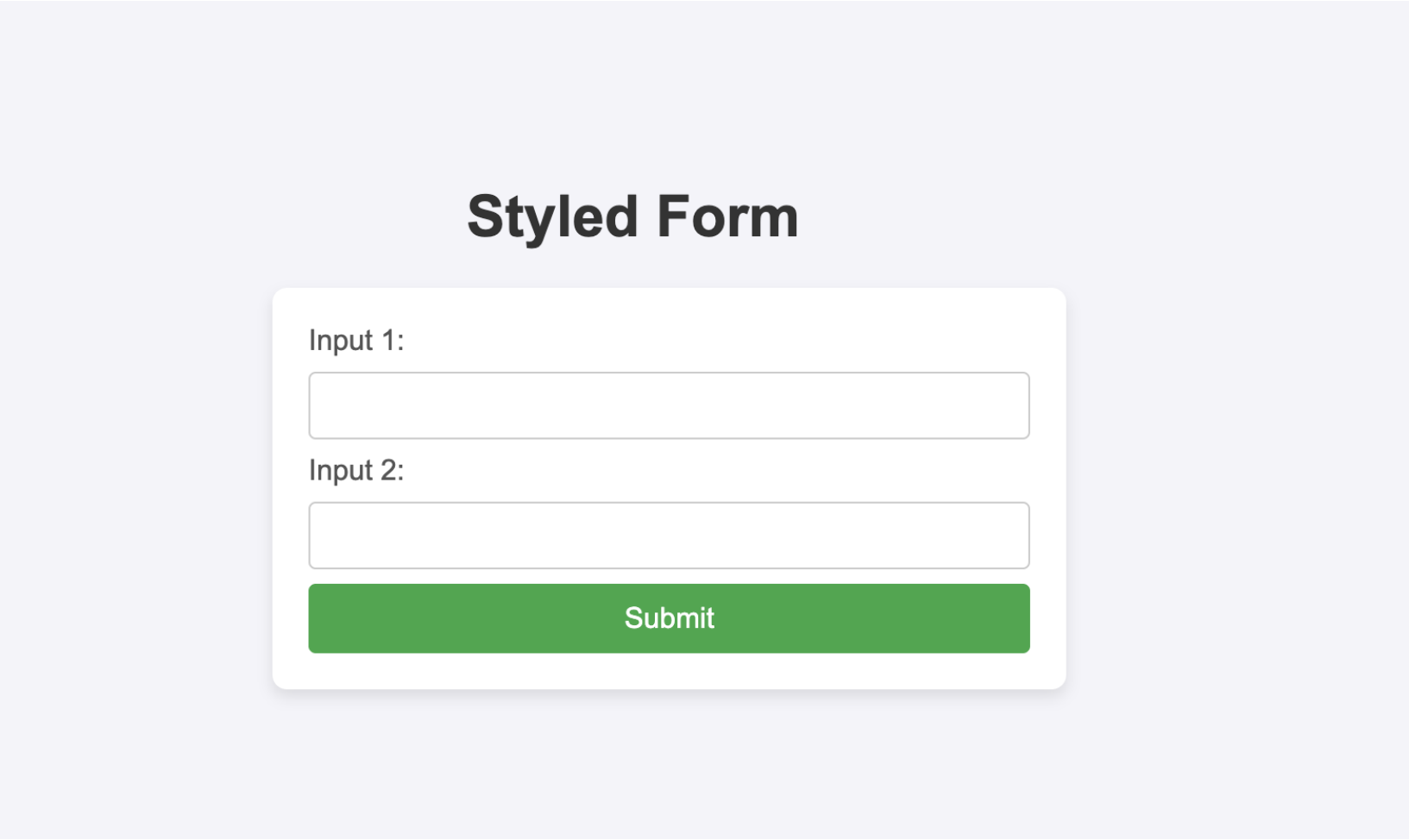
```css
button {
    background-color: #28a745;
    color: white;
    padding: 10px 15px;
    border: none;
    border-radius: 4px;
    cursor: pointer;
    width: 100%;
    font-size: 16px;
}

button:hover {
    background-color: #218838;
}

.container {
    width: 100%;
    max-width: 400px;
    margin: 0 auto;
}
```

Introduction

❖ HTML + CSS



Introduction

❖ HTML + CSS + JS

```html
<body>
    <div class="container">
        <h1>Styled Form</h1>
        <form id="concatForm" action="javascript:void(0);">
            <label for="input1">First name:</label>
            <input type="text" id="input1" name="input1" required>
            <label for="input2">Last name:</label>
            <input type="text" id="input2" name="input2" required>
            <button type="submit">Submit</button>
        </form>
        <p id="result"></p>
    </div>
    <script src="index.js"></script>
</body>
```

```javascript
document.getElementById('concatForm').addEventListener('submit', function(event) {
    event.preventDefault();

    const input1 = document.getElementById('input1').value;
    const input2 = document.getElementById('input2').value;

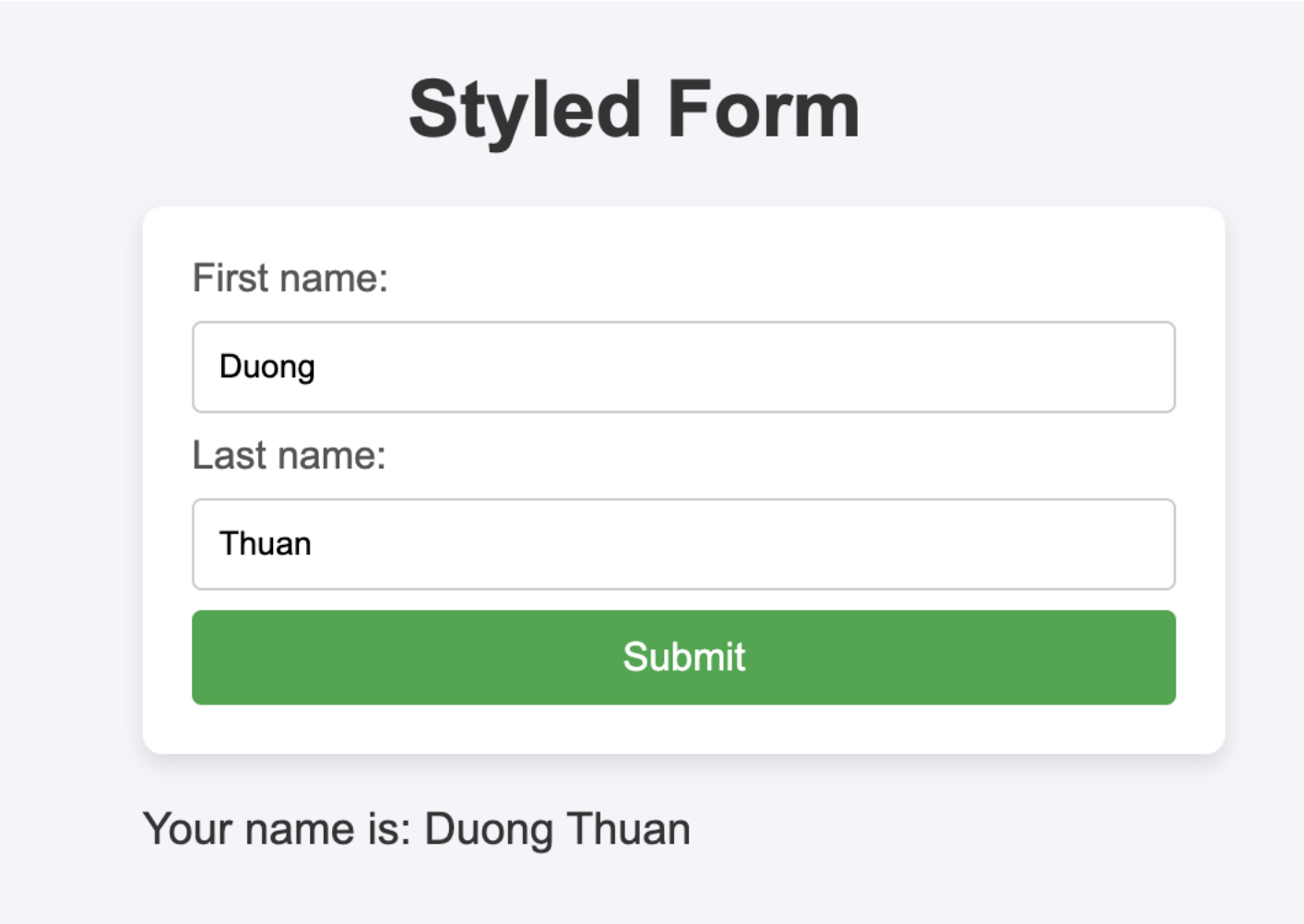
    const result = input1 + " " + input2;

    document.getElementById('result').textContent = "Your name is: " + result;
});

```

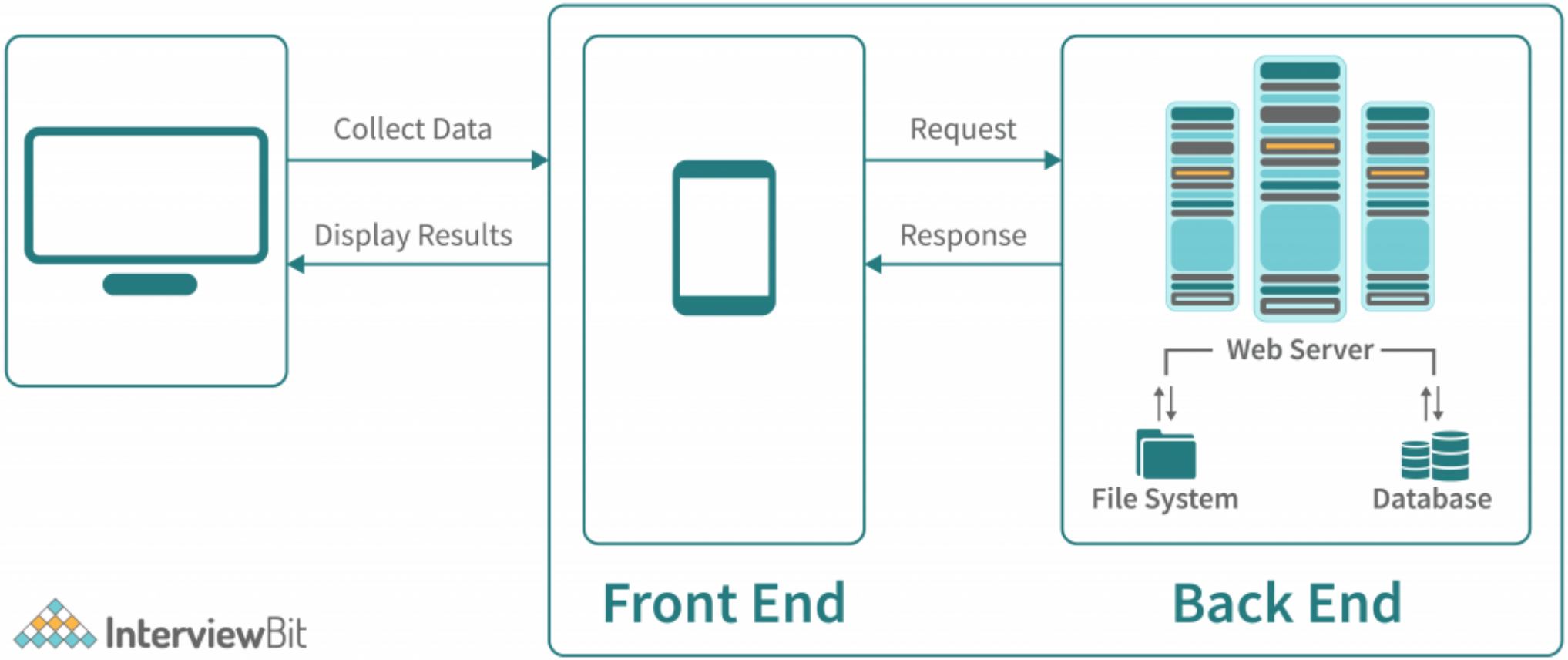
Introduction

- **HTML + CSS + JS**



Introduction

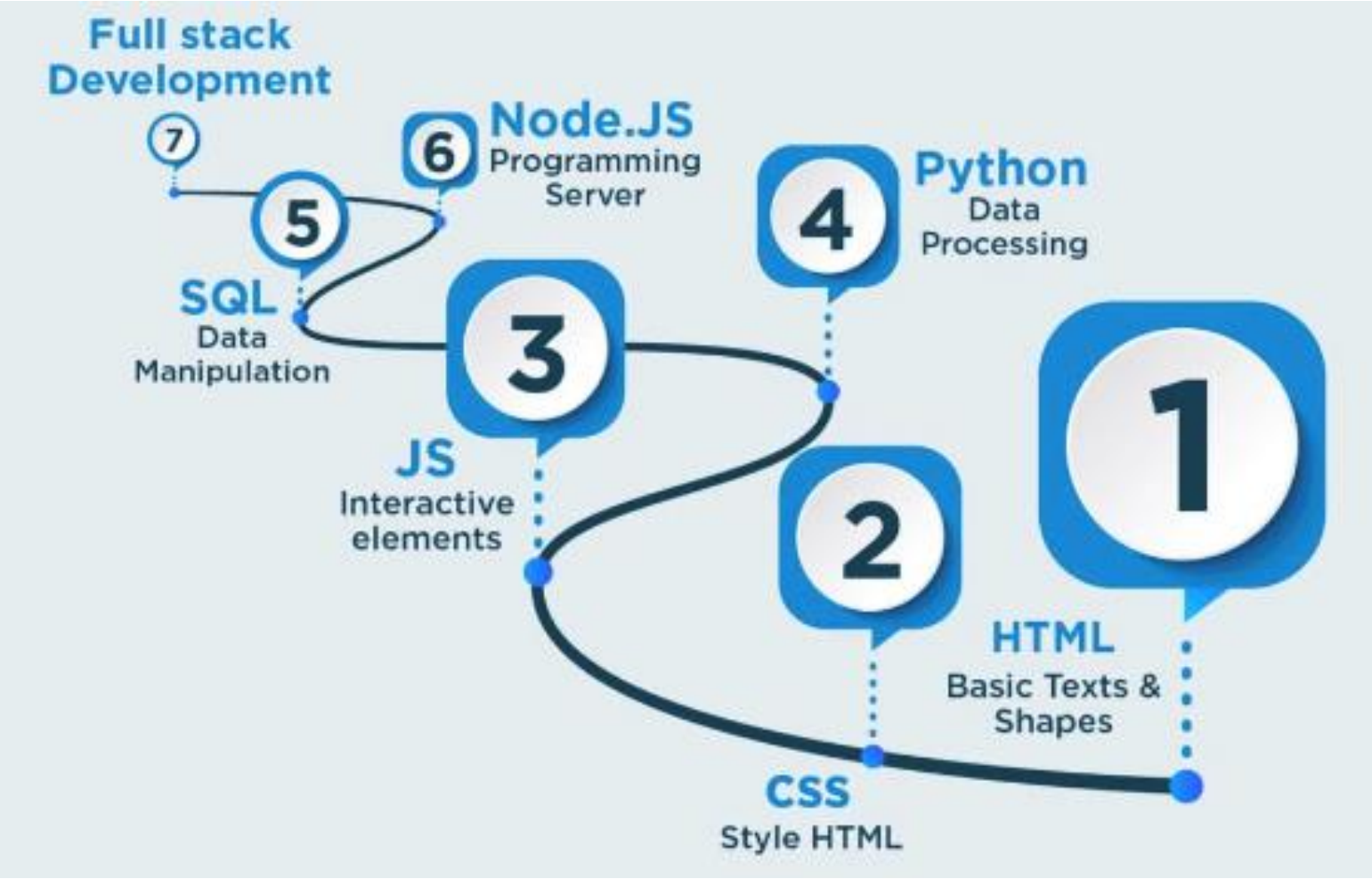
❖ Web Development



Ref: https://www.interviewbit.com/blog/web-application-architecture

Introduction

❖ Web Development

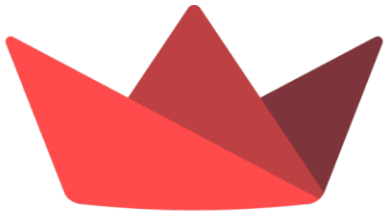


Introduction

❖ **No HTML - No CSS - No JavaScript**



gradio



Streamlit



Chainlit

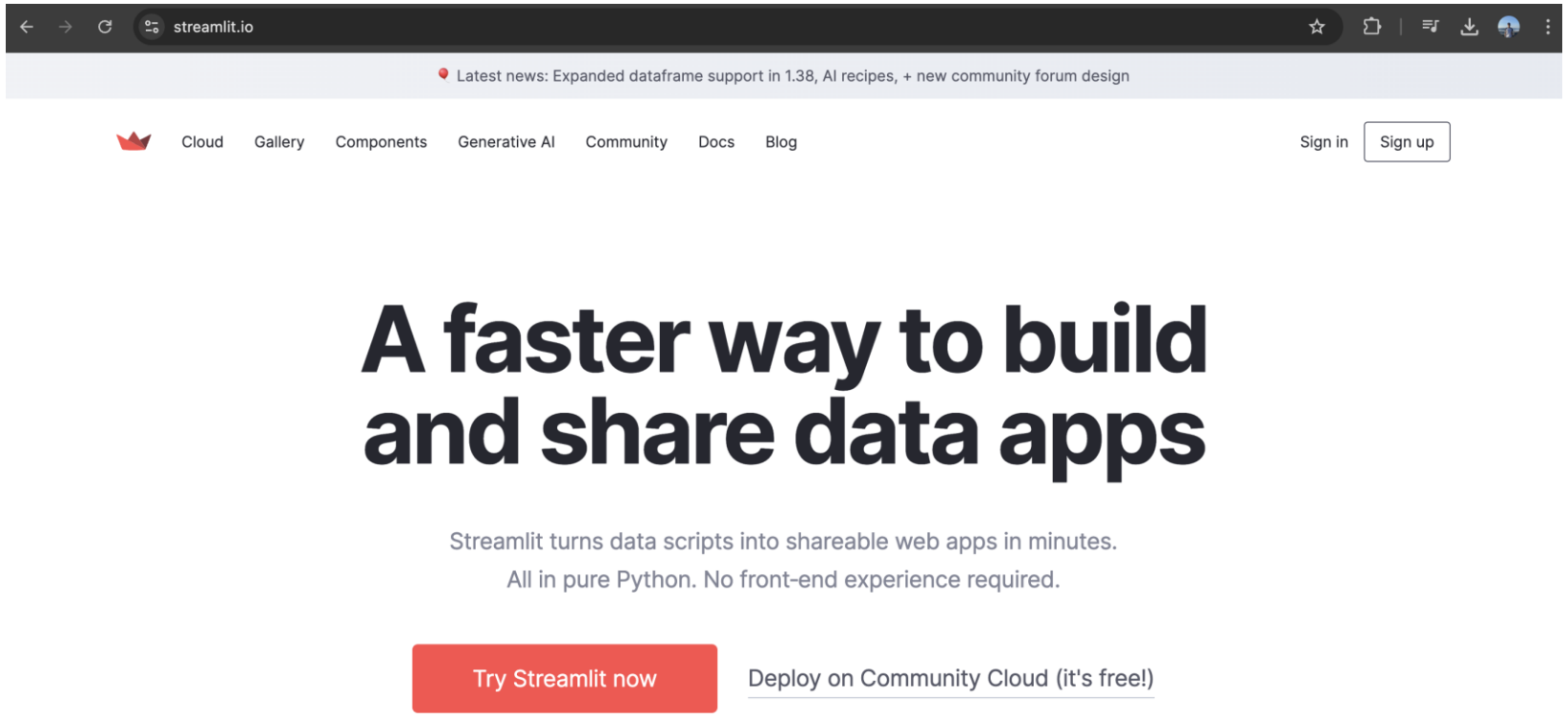


Streamlit

Streamlit

❖ Getting Started

Streamlit's open-source app framework is a breeze to get started with.

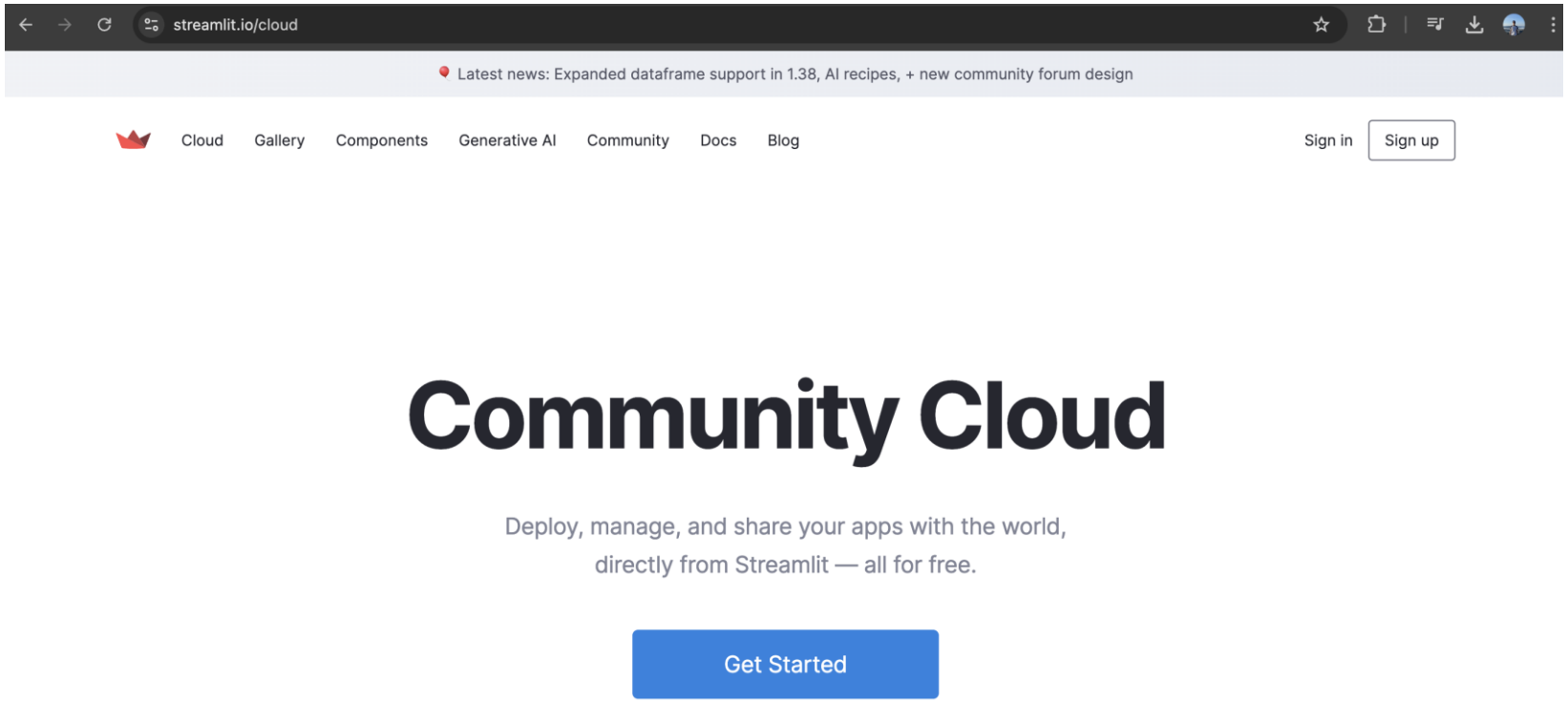


Ref: https://streamlit.io

Streamlit

❖ Getting Started

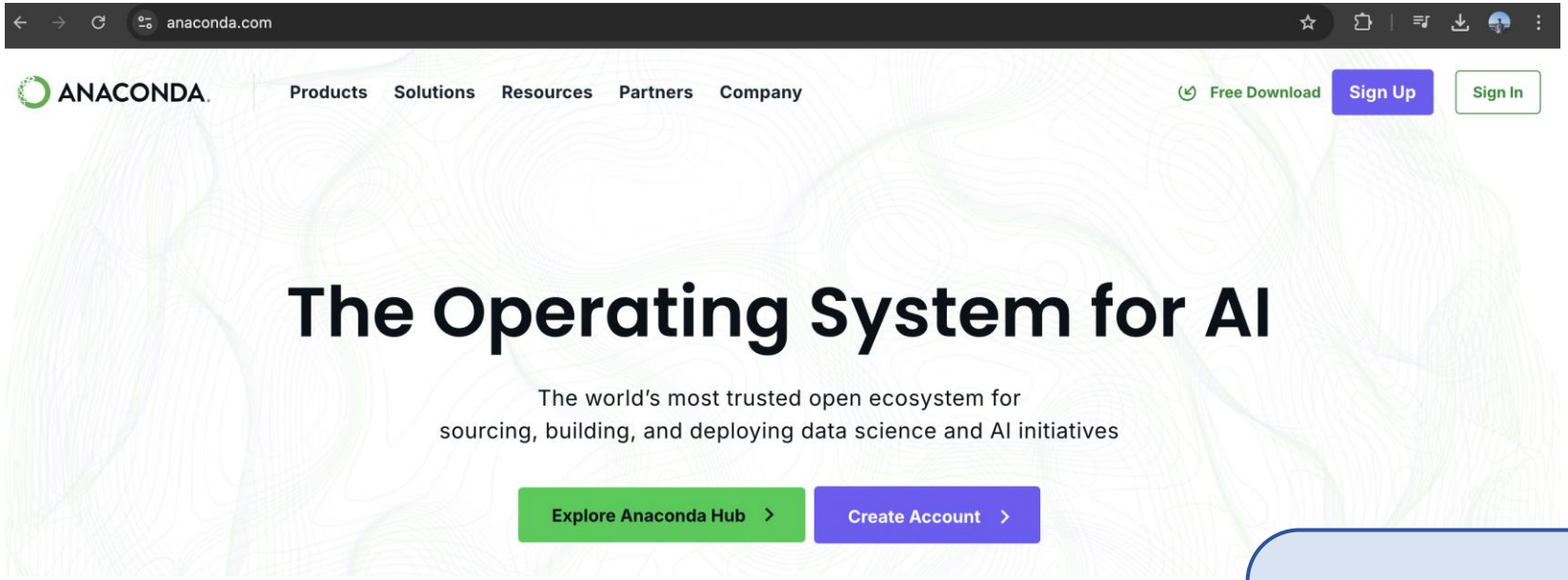
Streamlit also supports deploying applications built with Streamlit to the cloud easily and quickly.



Ref: https://streamlit.io

Streamlit

❖ Installation

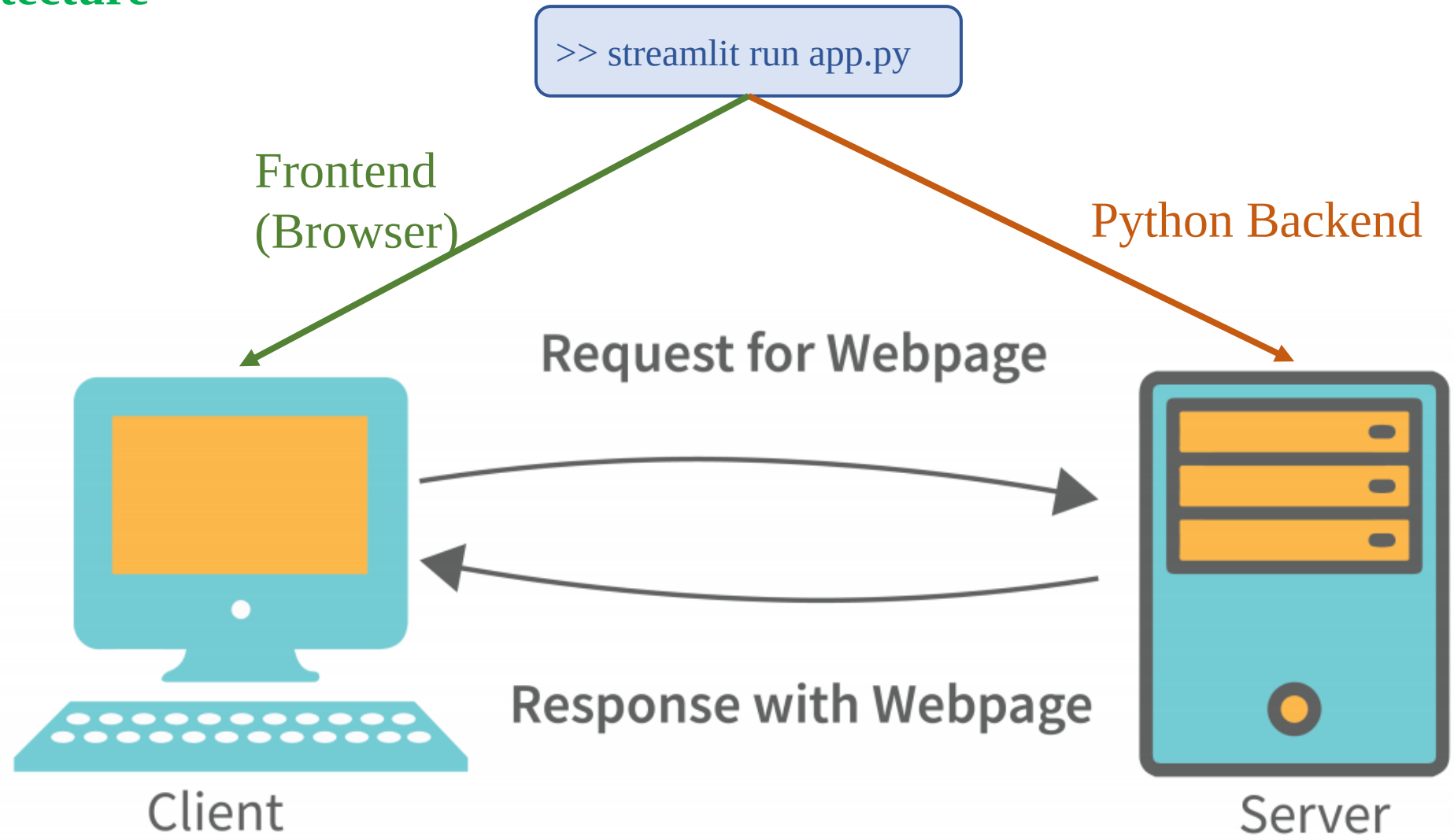



```
>> conda create –n streamlit_env python=3.11.9
>> conda activate streamlit_env
>> pip3 install streamlit==1.39.0
```

Ref: https://www.anaconda.com

Streamlit

❖ Architecture



Streamlit

❖ First app

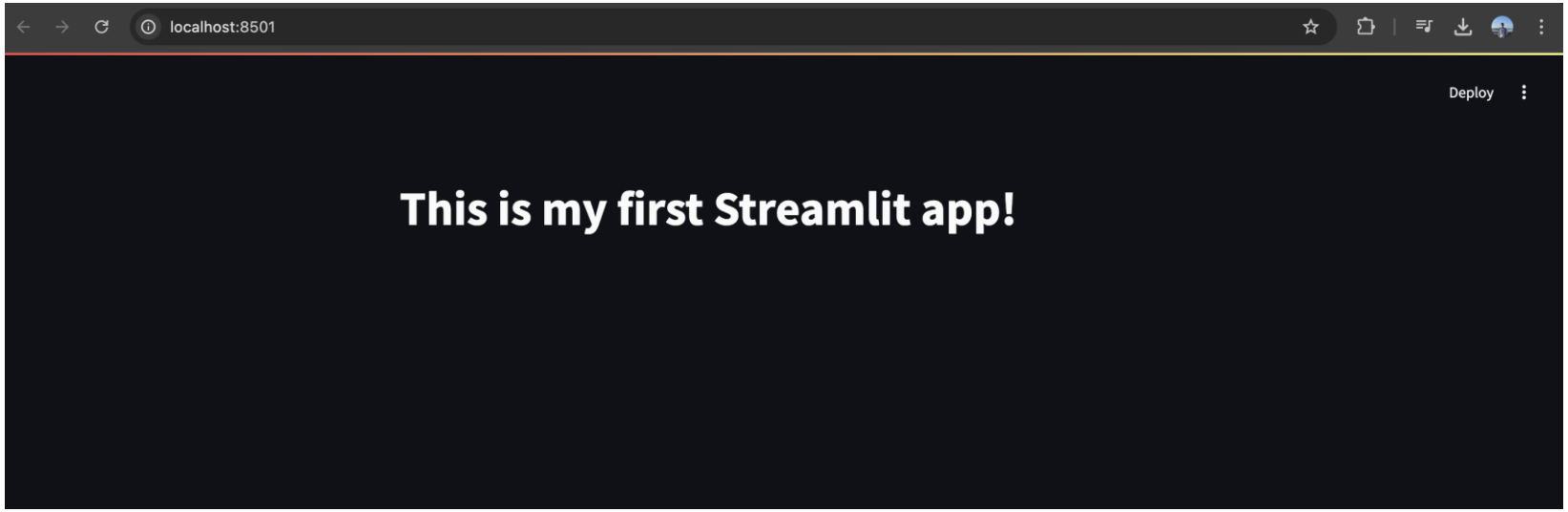
```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")
```

```
PROBLEMS 6  OUTPUT  TERMINAL  PORTS  JUPYTER  DEBUG CONSOLE
(aio-mlops-w1) thuannd@MacNaN streamlit % streamlit run app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.1.5:8501
```



Text element

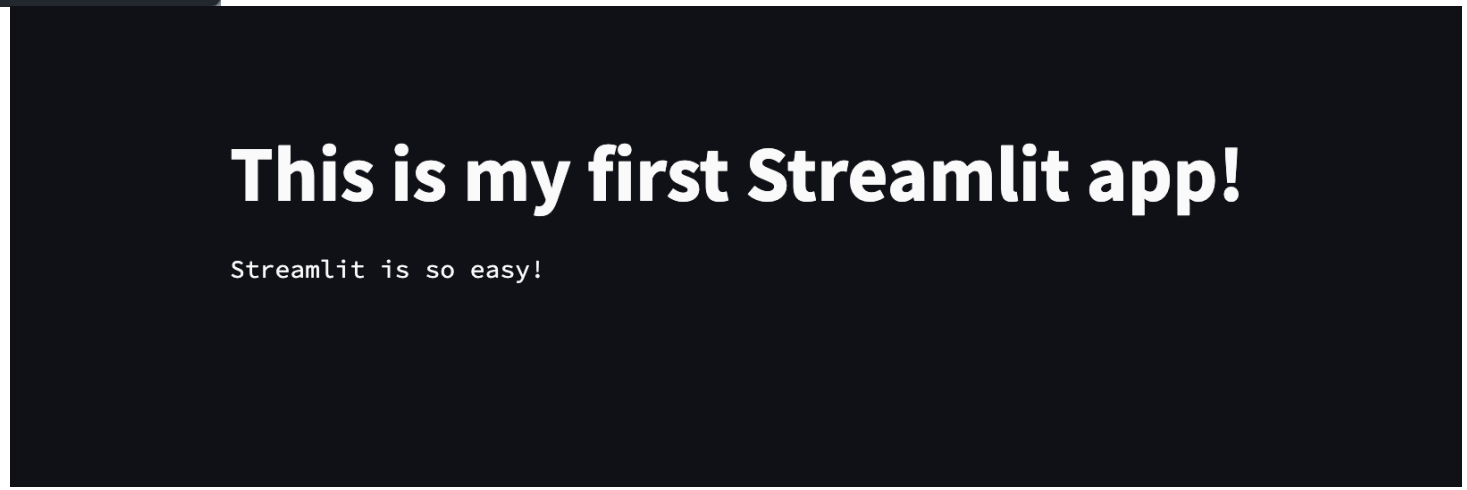
Text element

❖ st.text

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

st.text("Streamlit is so easy!")
```

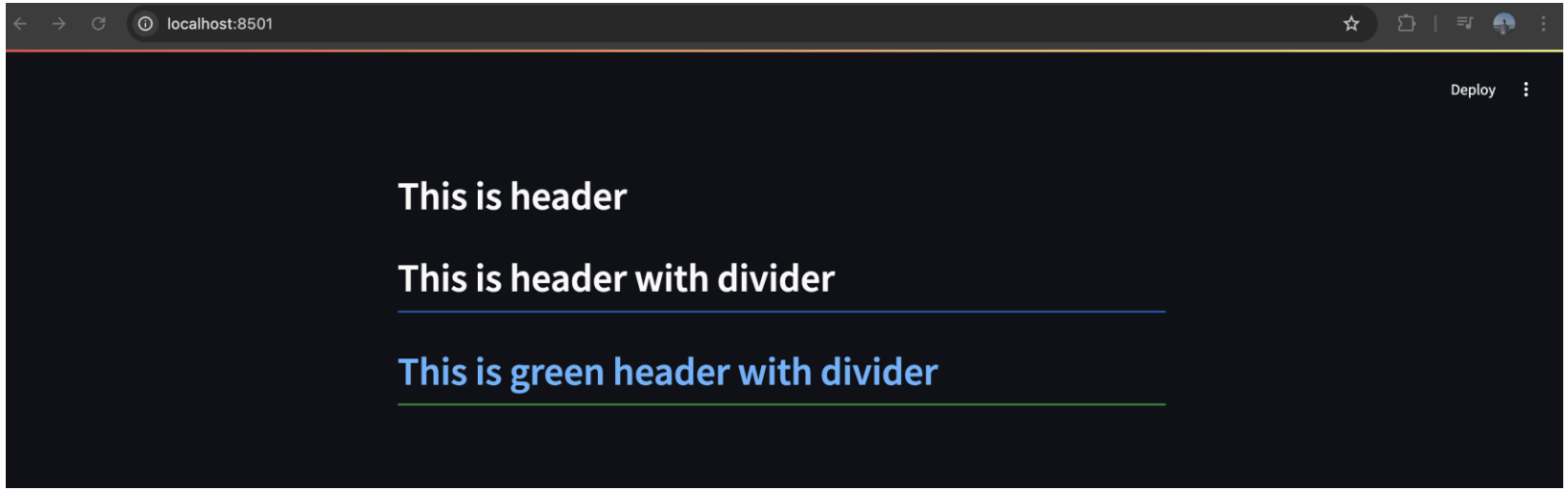


Text element

❖ st.header

```python
import streamlit as st

st.header("This is header")
st.header("This is header with divider", divider=True)
st.header(":blue[This is green header with divider]", divider=True)
```



Text element

❖ st.markdown

```python
import streamlit as st

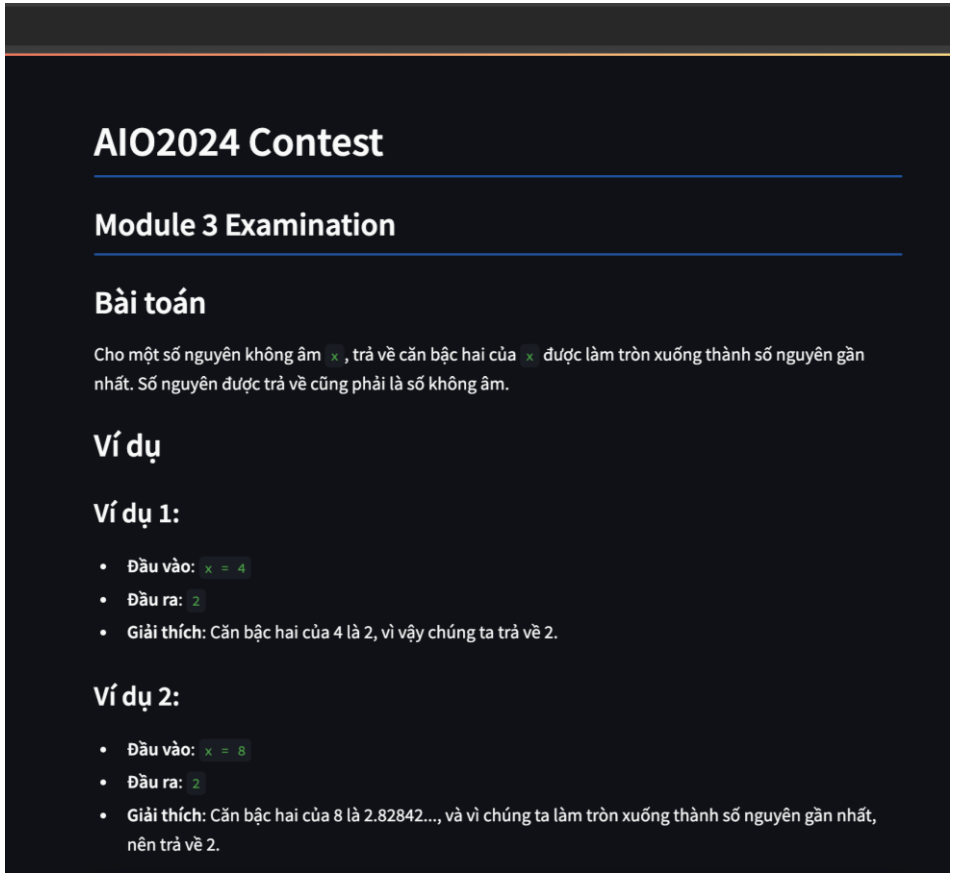
st.header("AIO2024 Contest", divider=True)
st.subheader("Module 3 Examination", divider=True)

problem = """
### **Bài toán**
Cho một số nguyên không âm `x`, trả về căn bậc hai của `x` được làm tròn xuống thành số nguyên gần nhất. Số nguyên được t

### **Ví dụ**
#### Ví dụ 1:
- **Đầu vào**: `x = 4`
- **Đầu ra**: `2`
- **Giải thích**: Căn bậc hai của 4 là 2, vì vậy chúng ta trả về 2.

#### Ví dụ 2:
- **Đầu vào**: `x = 8`
- **Đầu ra**: `2`
- **Giải thích**: Căn bậc hai của 8 là 2.82842..., và vì chúng ta làm tròn xuống thành số nguyên gần nhất, nên trả về 2.
"""

st.markdown(problem)
```



Text element

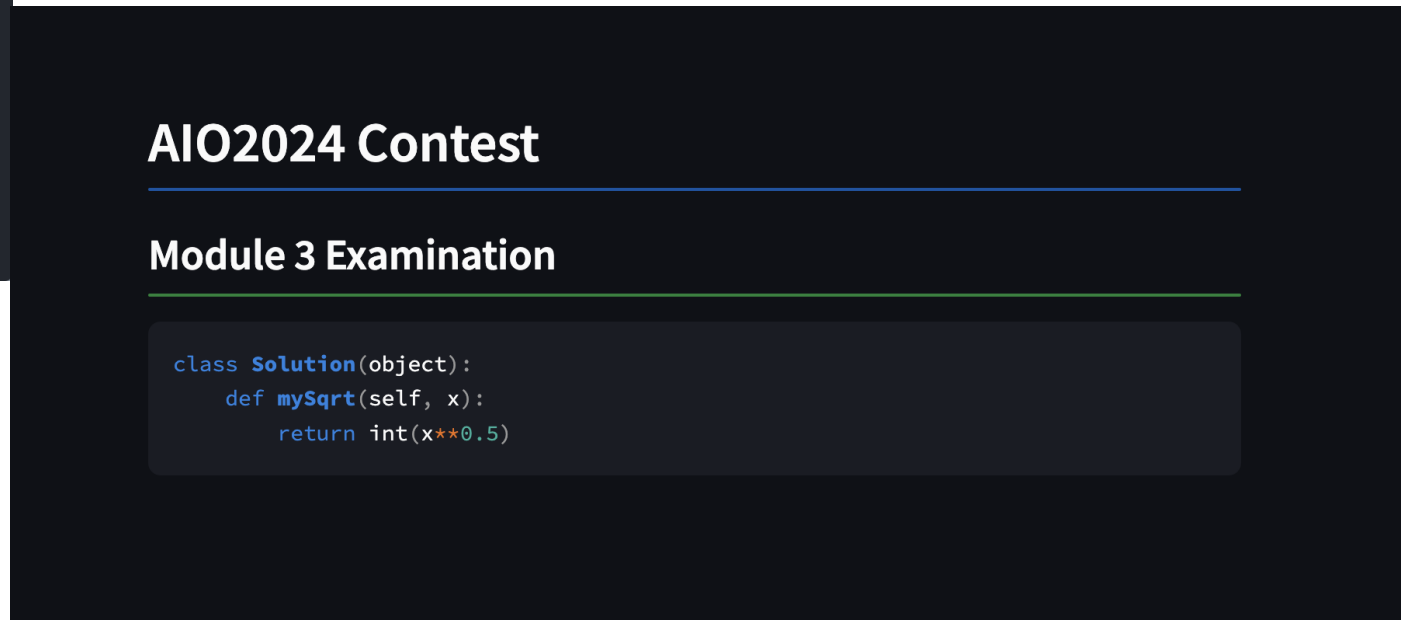
❖ st.code

```python
import streamlit as st

st.header("AIO2024 Contest", divider=True)
st.subheader("Module 3 Examination", divider=True)

code = """
class Solution(object):
    def mySqrt(self, x):
        return int(x**0.5)
"""

st.code(code, language="python")
```



Input element

Input element

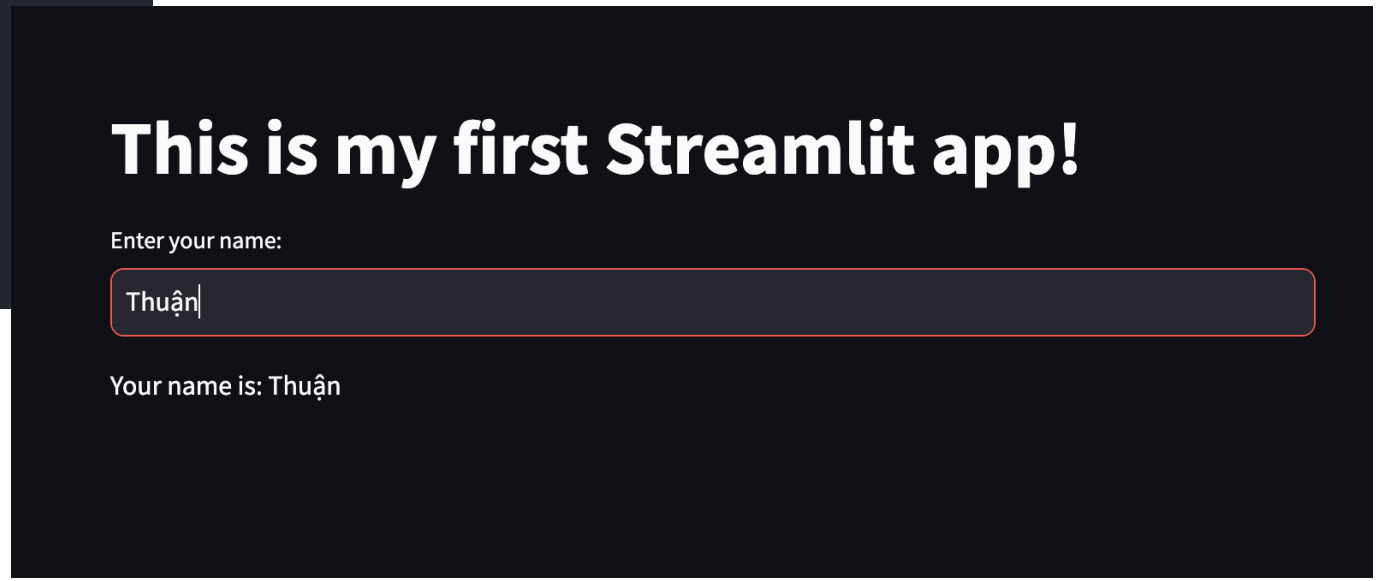
❖ st.text_input

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

name = st.text_input(
    label="Enter your name: ",
)

st.write("Your name is: ", name)
```



Input element

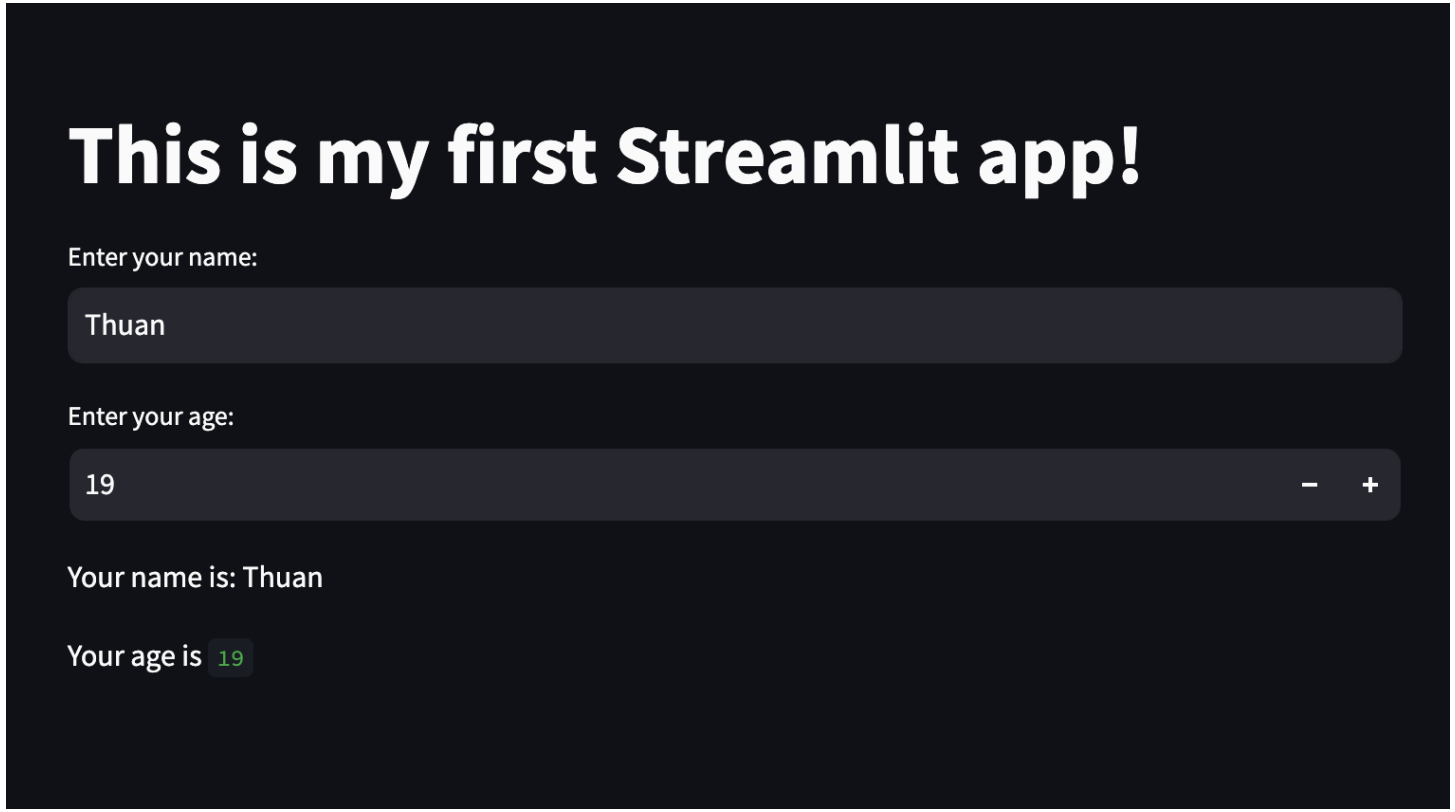
❖ st.number_input

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

name = st.text_input(
    label="Enter your name: ",
    value="",
)
age = st.number_input(
    "Enter your age: ",
    value=0,
    step=1,
    format="%d"
)

st.write("Your name is: ", name)
st.write("Your age is ", age)
```



Input element

❖ st.slider

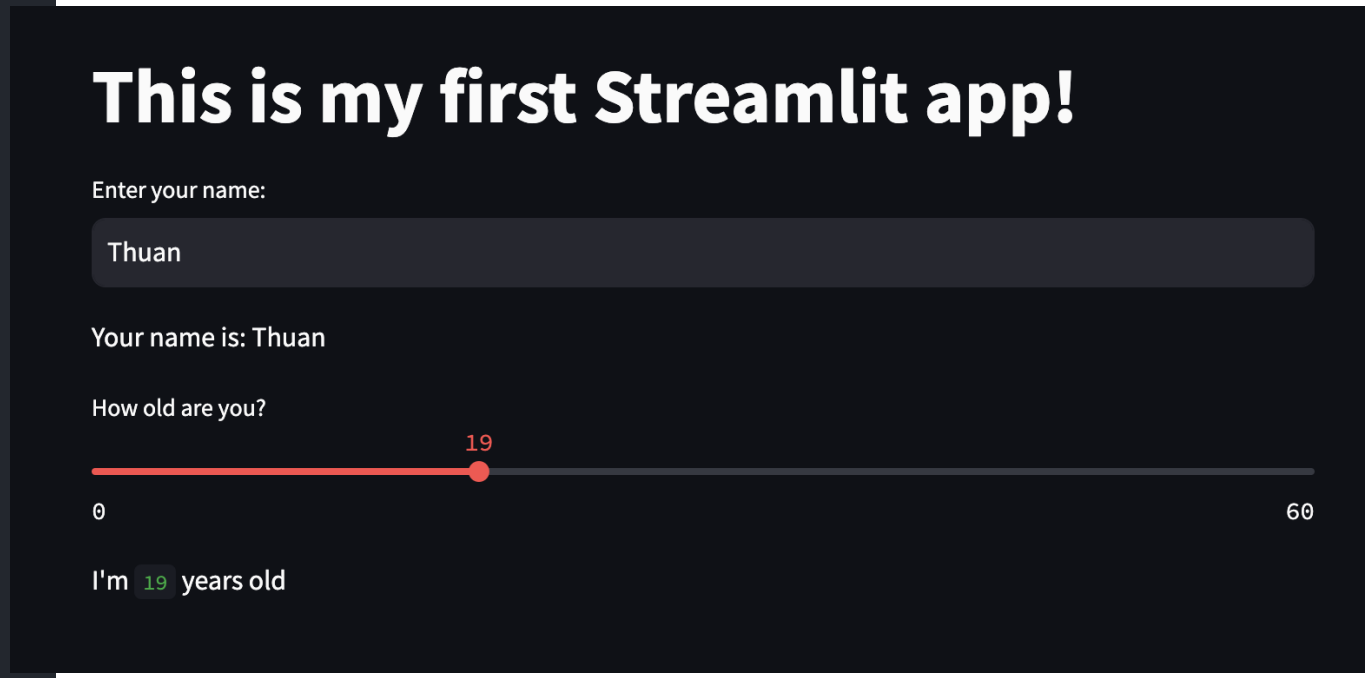
```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

name = st.text_input(
    label="Enter your name: ",
    value="",
)
st.write("Your name is: ", name)


age = st.slider("How old are you?", 0, 60, 20)
st.write("I'm ", age, "years old")

```



Input element

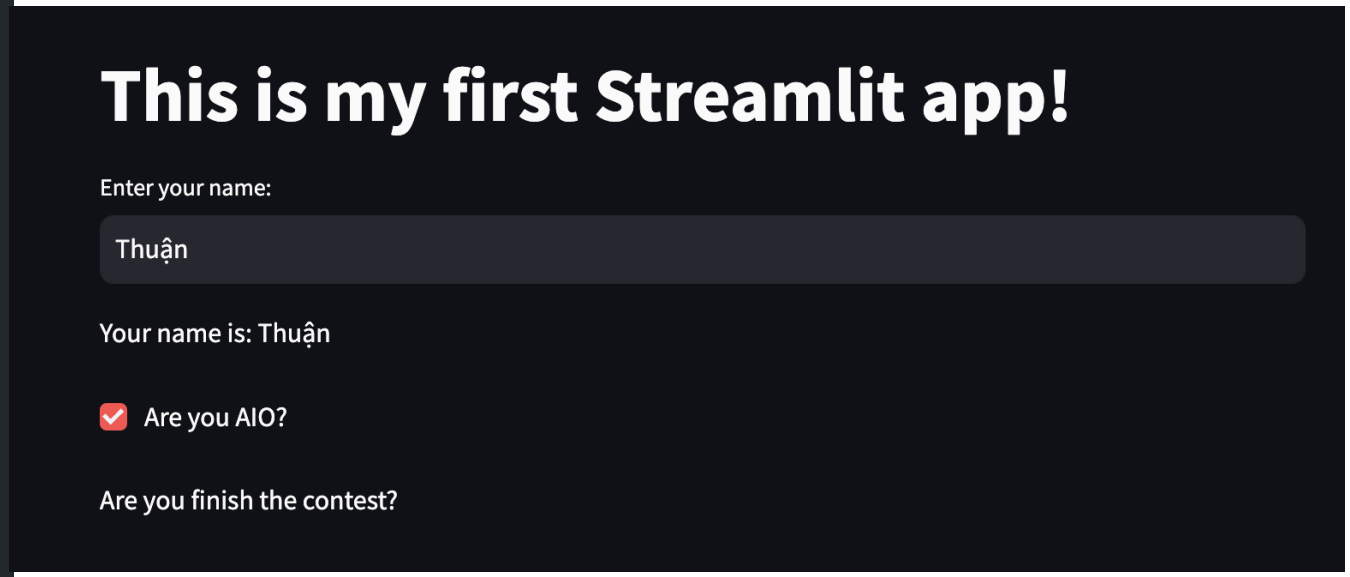
❖ st.checkbox

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

name = st.text_input(
    label="Enter your name: ",
    value="",
)
st.write("Your name is: ", name)

is_aio = st.checkbox("Are you AIO?")
if is_aio:
    st.write("Are you finish the contest?")
```



Input element

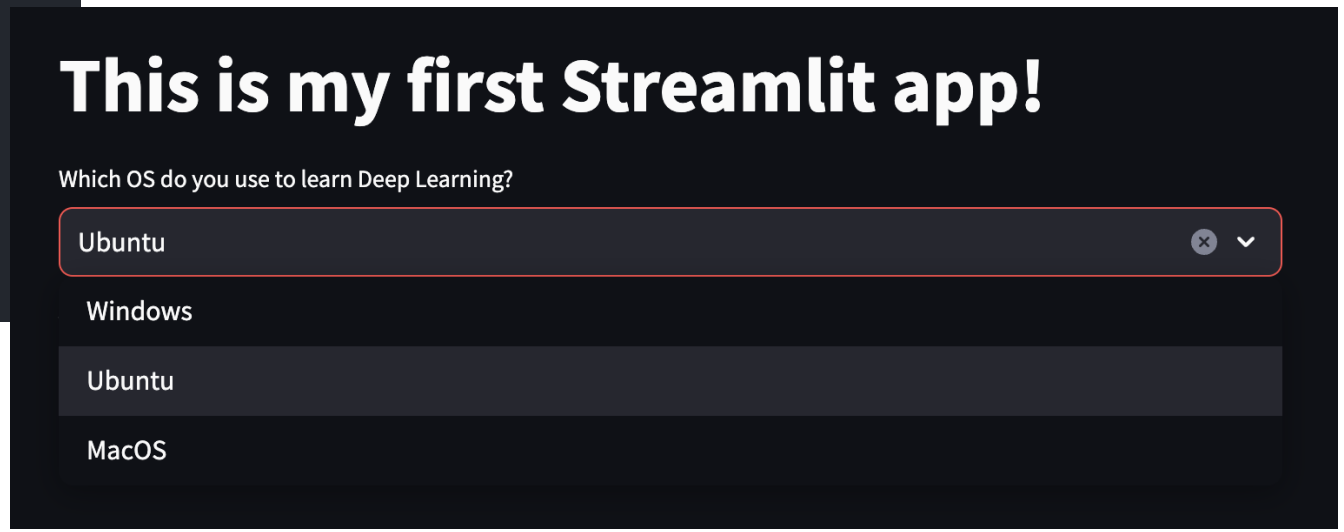
❖ st.selectbox

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

option = st.selectbox(
    "Which OS do you use to learn Deep Learning?",
    ("Windows", "Ubuntu", "MacOS"),
    index=None,
    placeholder="Select an OS...",
)

st.write("You choosed:", option)
```



Input element

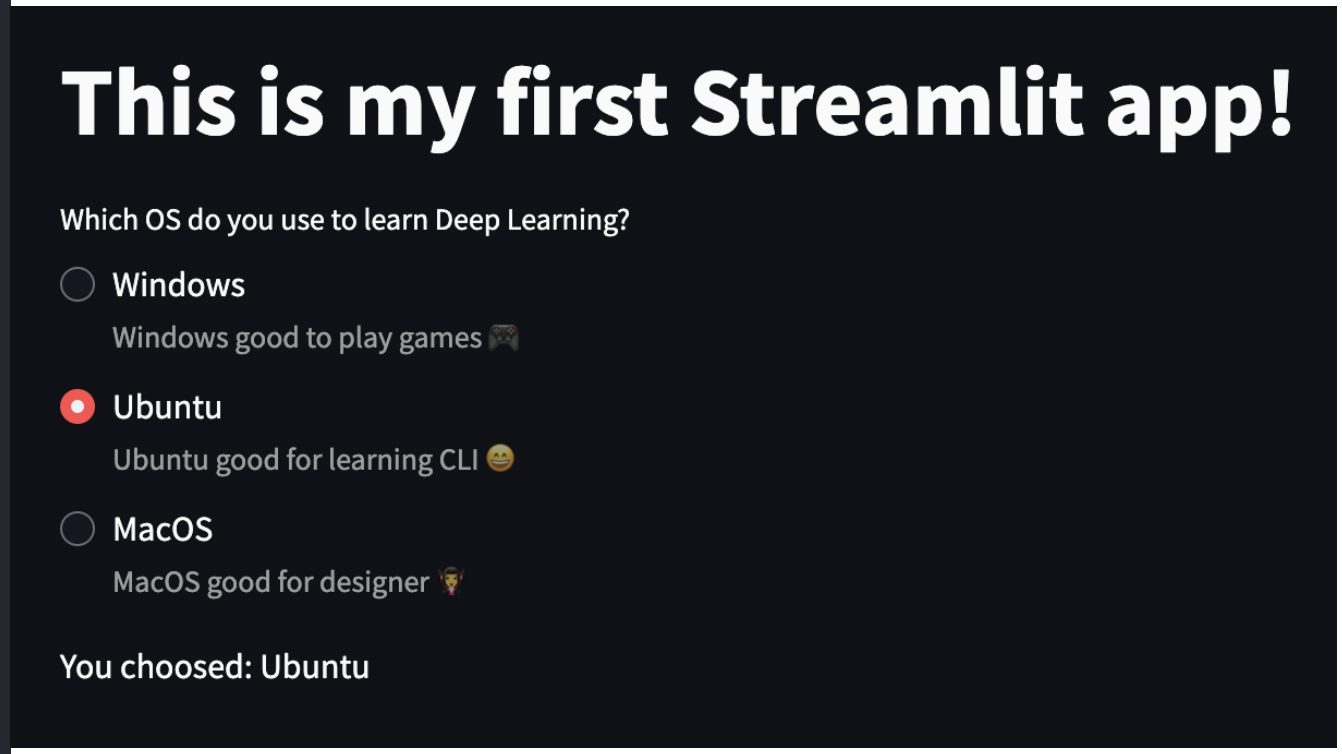
❖ st.radio

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

option = st.radio(
    "Which OS do you use to learn Deep Learning?",
    ["Windows", "Ubuntu", "MacOS"],
    captions=[
        "Windows good to play games :video_game:",
        "Ubuntu good for learning CLI :smile:",
        "MacOS good for designer :vampire:",
    ]
)

st.write("You choosed:", option)
```



Input element

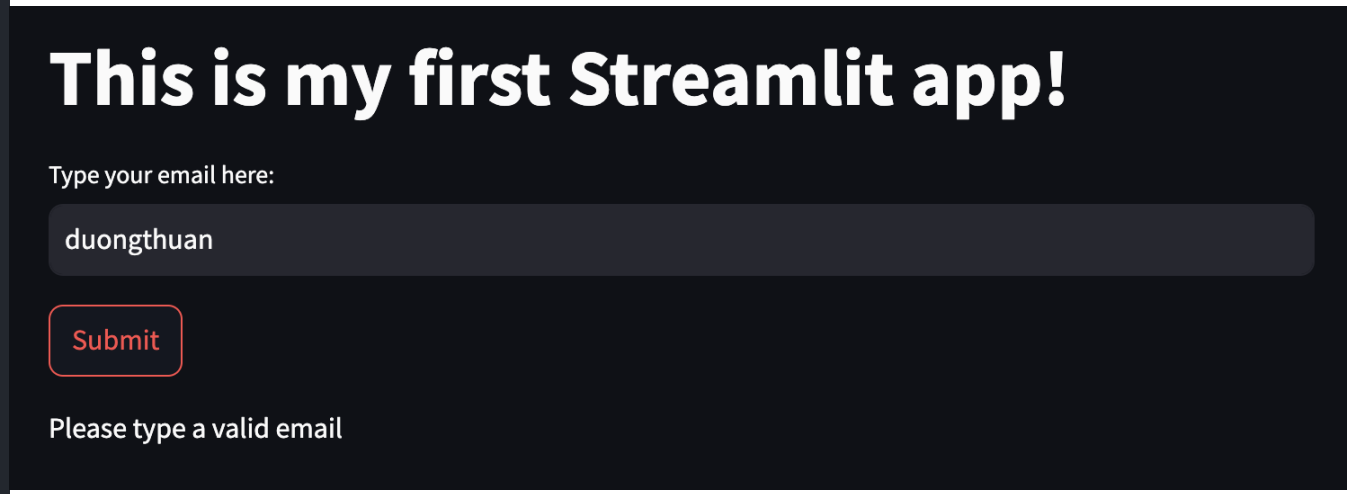
❖ st.button

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

email = st.text_input(
    label="Type your email here: "
)

if st.button("Submit"):
    if "@" not in email:
        st.write("Please type a valid email")
    else:
        st.write("Submitted")
```



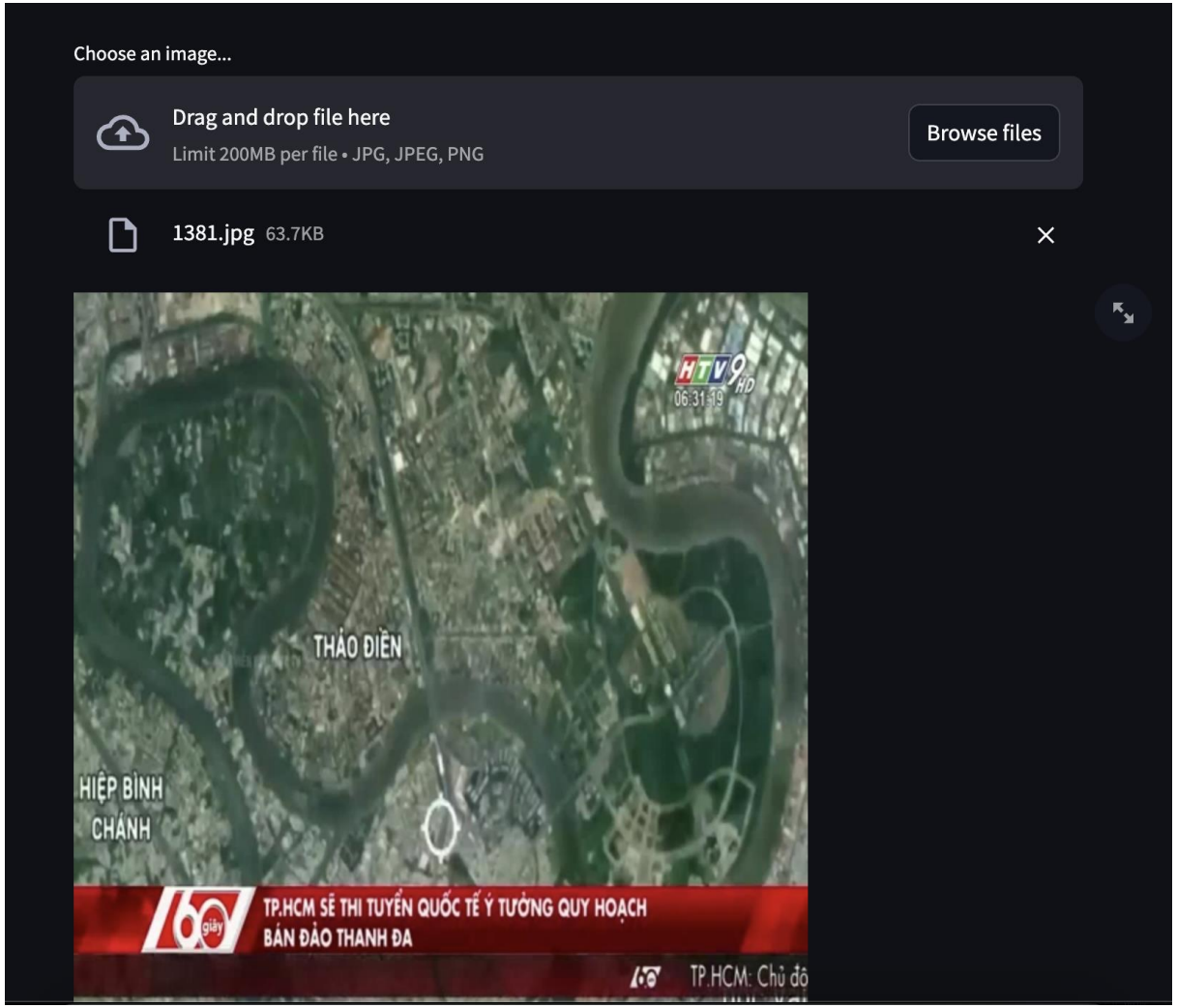
Input element

❖ st.file_uploader

```python
# app.py
import streamlit as st
from PIL import Image

uploaded_file = st.file_uploader("Choose an image...",
                                 type=["jpg", "jpeg", "png"])

if uploaded_file is not None:
    img = Image.open(uploaded_file)
    img = img.resize((224, 224))
    st.image(img, caption="Uploaded Image.")
```



Media element

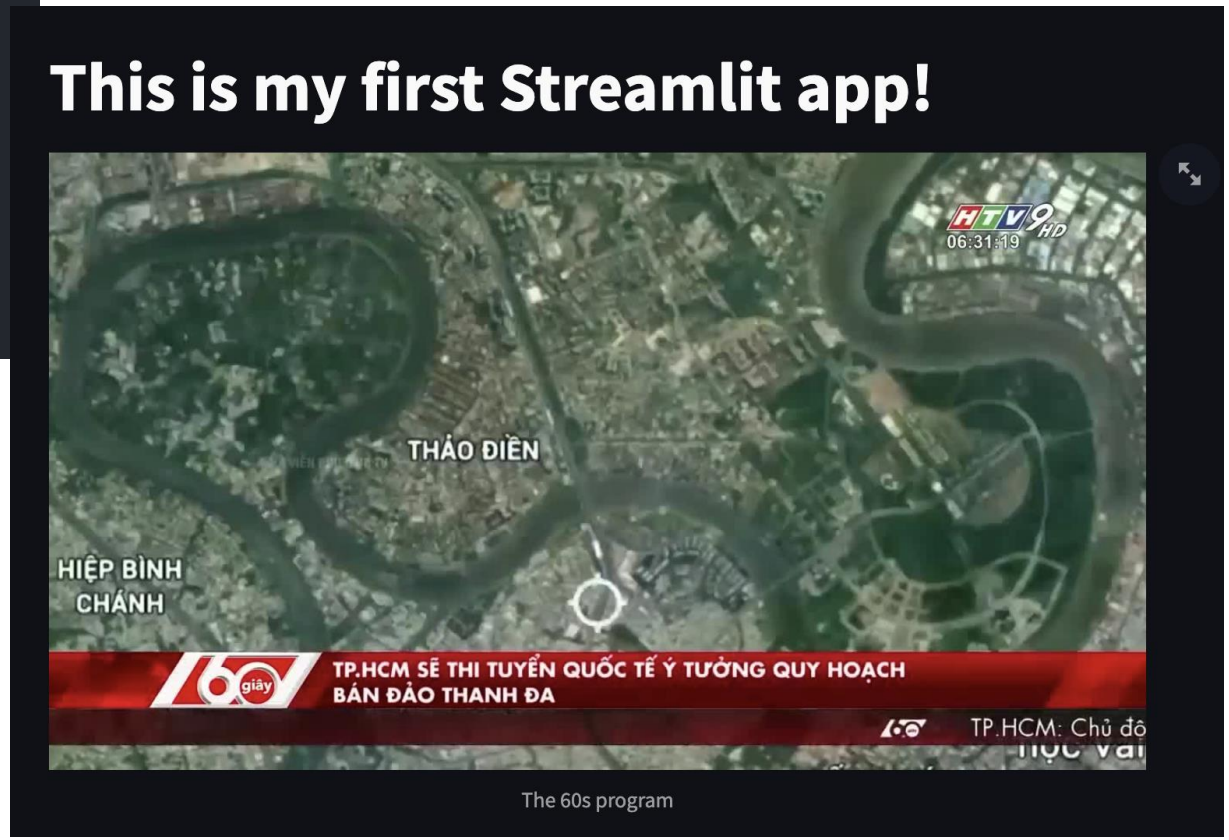
Media element

❖ st.image

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

st.image("./60s.jpg", caption="The 60s program")
```



Media element

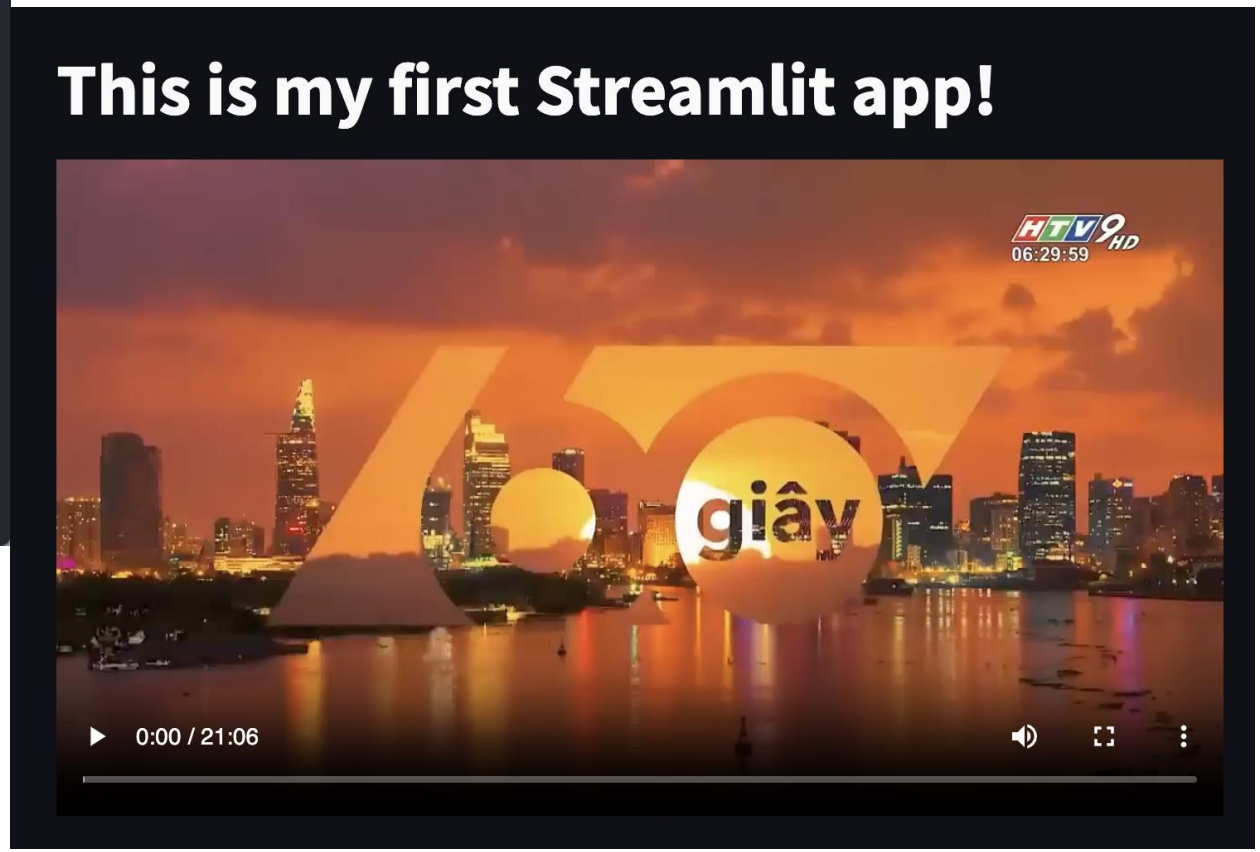
❖ st.video

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

video_file = open("./L01_V001.mp4", "rb")
video_bytes = video_file.read()

st.video(video_bytes)
```



Layout element

Layout element

❖ st.columns

```python
# app.py
import streamlit as st

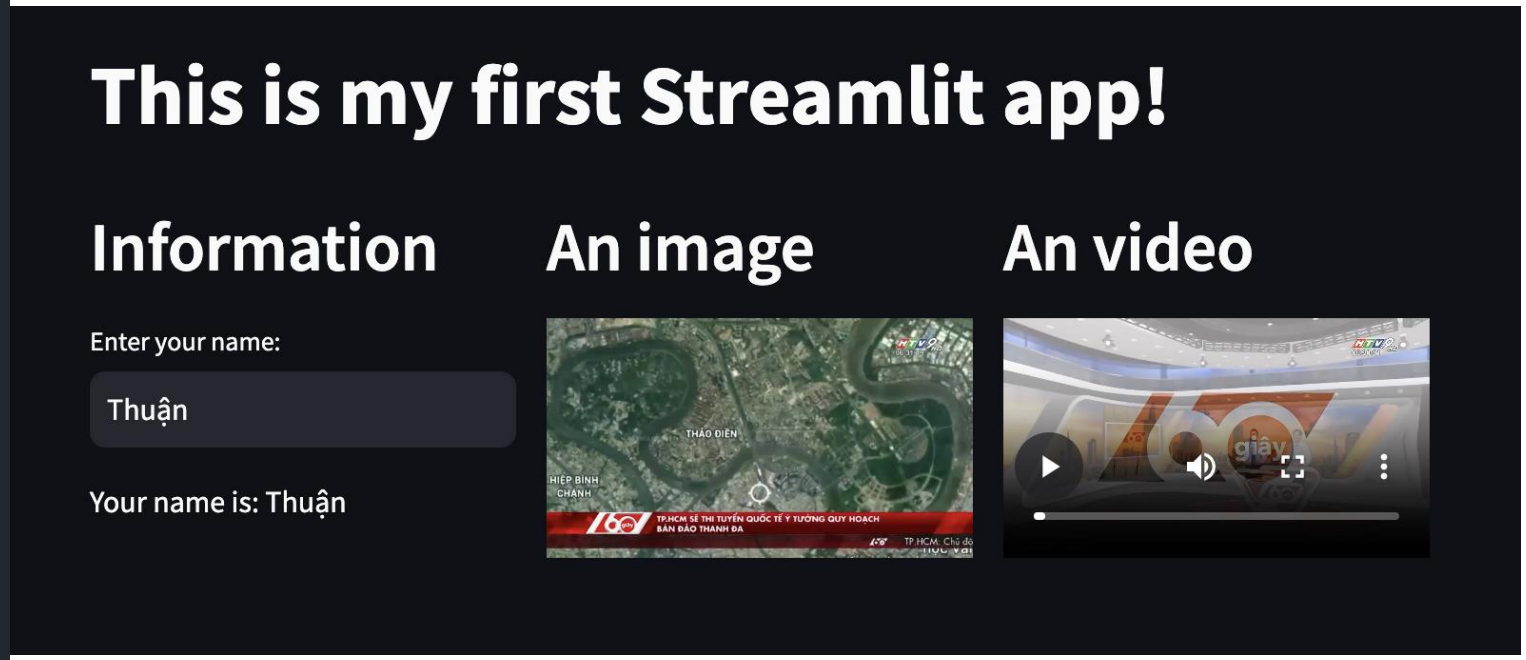
st.title("This is my first Streamlit app!")

col1, col2, col3 = st.columns(3)

with col1:
    st.header("Information")
    name = st.text_input(
        label="Enter your name: ",
    )
    st.write("Your name is:", name)

with col2:
    st.header("An image")
    st.image("./60s.jpg")

with col3:
    st.header("An video")
    video_file = open("./L01_V001.mp4", "rb")
    video_bytes = video_file.read()
    st.video(video_bytes)
```



Layout element

❖ st.container

```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

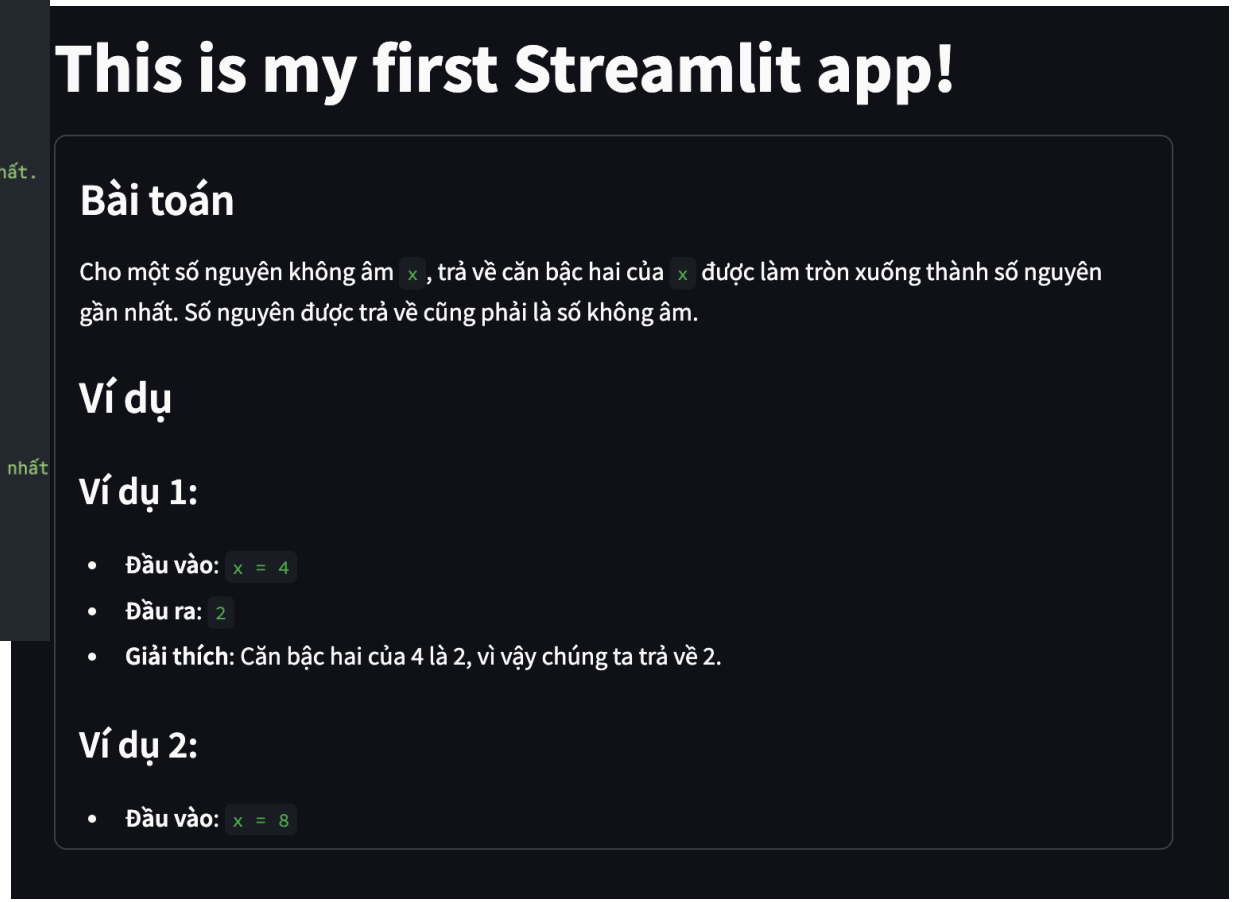
problem = """
### **Bài toán**
Cho một số nguyên không âm `x`, trả về căn bậc hai của `x` được làm tròn xuống thành số nguyên gần nhất.

### **Ví dụ**
#### Ví dụ 1:
- **Đầu vào**: `x = 4`
- **Đầu ra**: `2`
- **Giải thích**: Căn bậc hai của 4 là 2, vì vậy chúng ta trả về 2.

#### Ví dụ 2:
- **Đầu vào**: `x = 8`
- **Đầu ra**: `2`
- **Giải thích**: Căn bậc hai của 8 là 2.82842..., và vì chúng ta làm tròn xuống thành số nguyên gần nhất
"""

with st.container(height=450):
    st.markdown(problem)

```



Status element

Status element

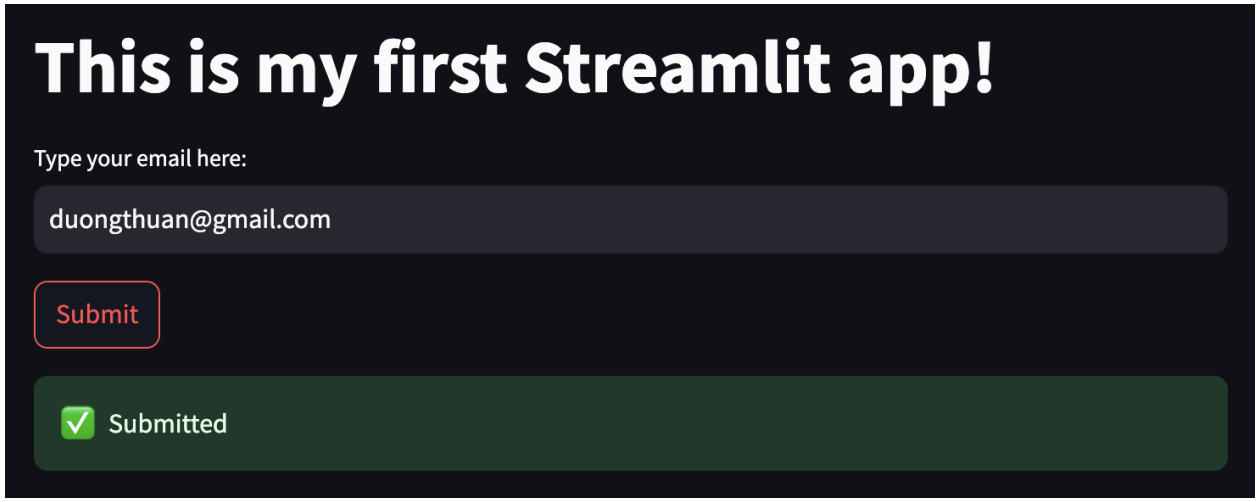
❖ st.error / st.success

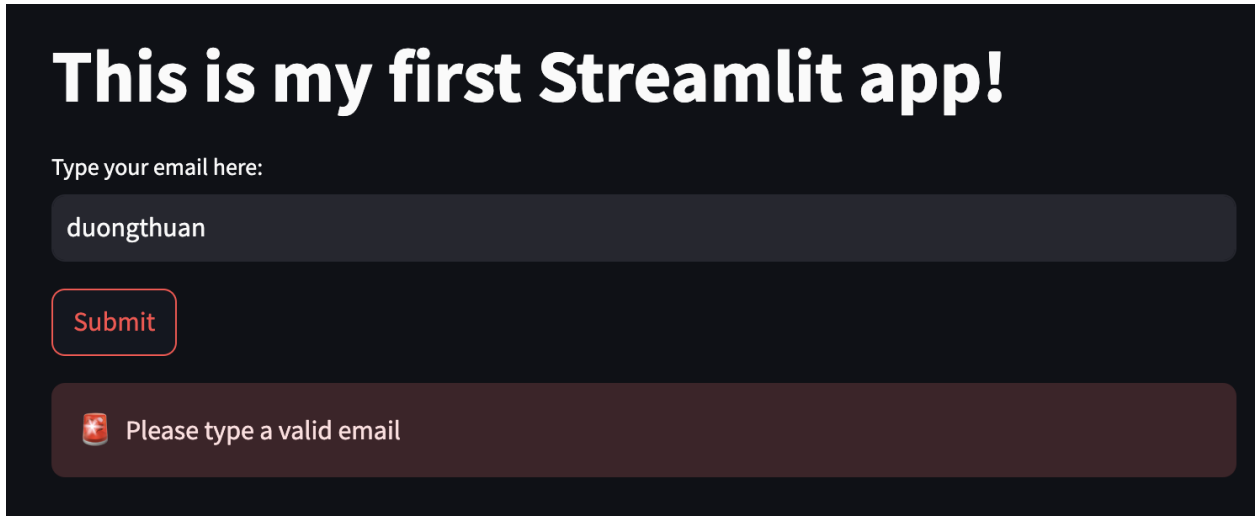
```python
# app.py
import streamlit as st

st.title("This is my first Streamlit app!")

email = st.text_input(
    label="Type your email here: "
)

if st.button("Submit"):
    if "@" not in email:
        st.error("Please type a valid email",
                 icon="🚨")
    else:
        st.success("Submitted",
                   icon="✅")
```





Status element

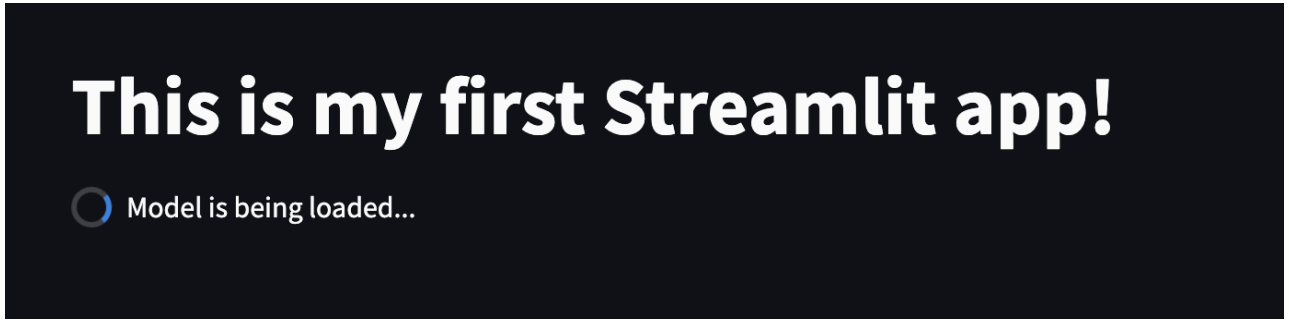
❖ st.spinner

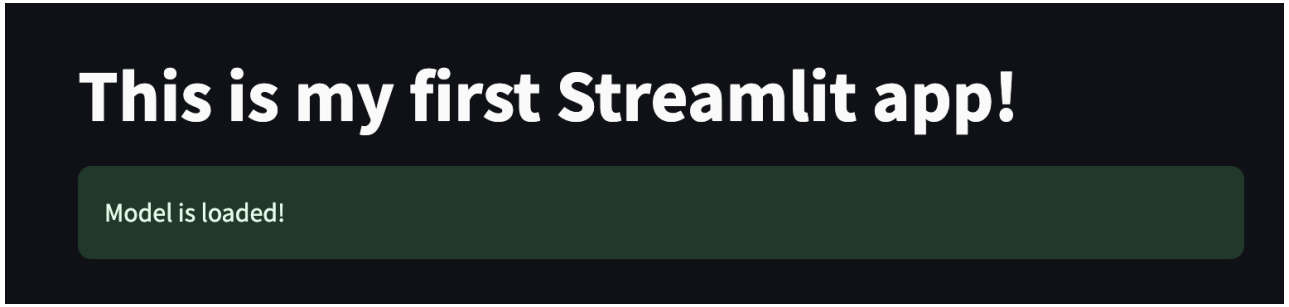
```python
# app.py
import time
import random
import streamlit as st

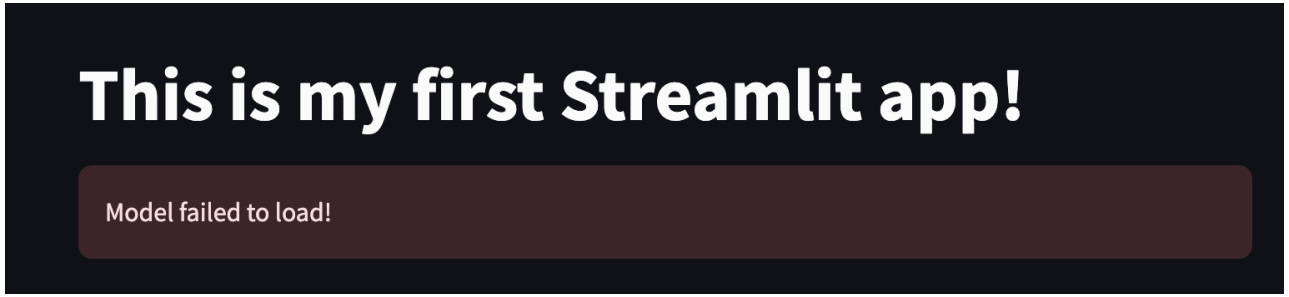
st.title("This is my first Streamlit app!")

def load_model():
    with st.spinner('Model is being loaded...'):
        time.sleep(3)
    is_loaded = random.choice([True, False])
    return is_loaded

model = load_model()
if model:
    st.success("Model is loaded!")
else:
    st.error("Model failed to load!")
```



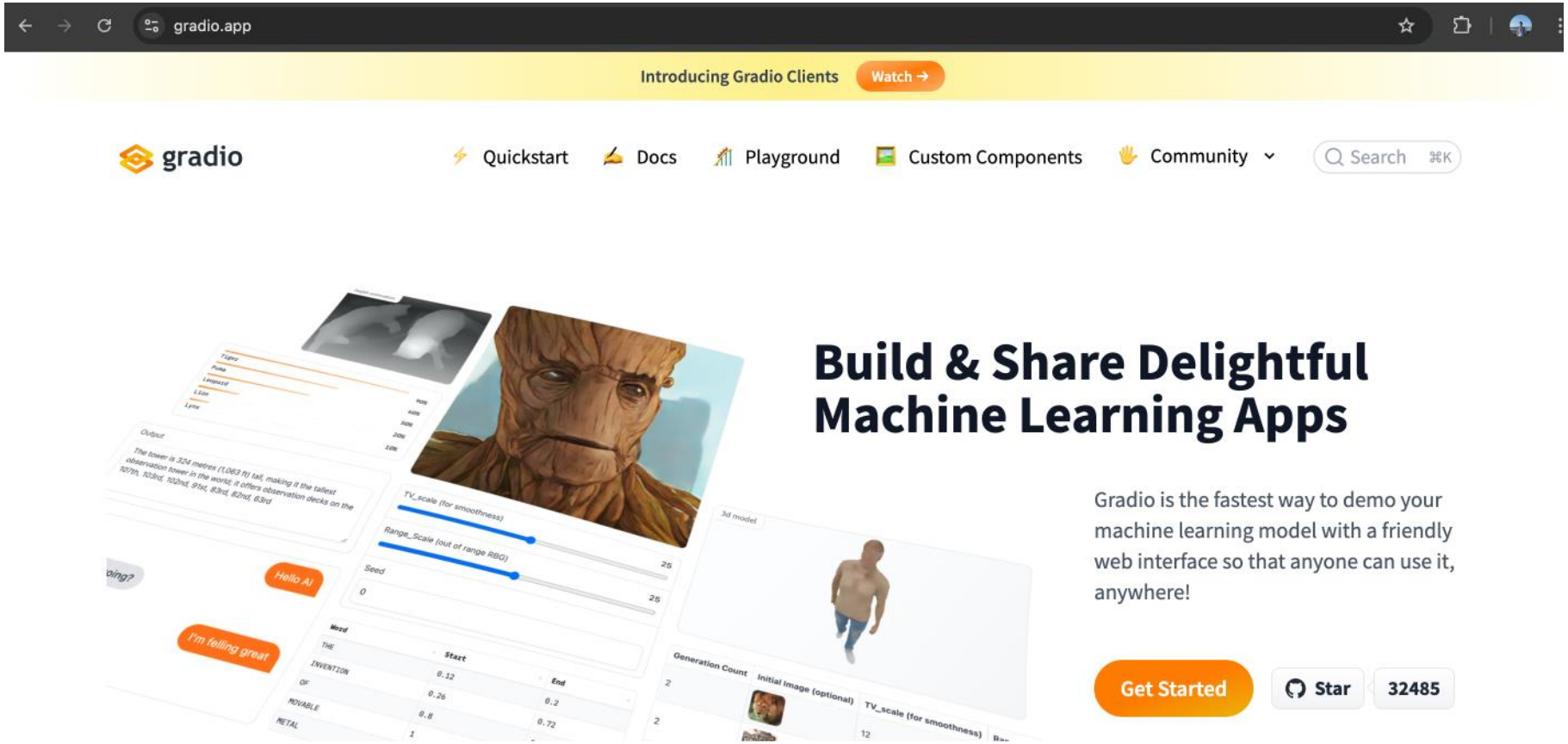




Gradio

Gradio

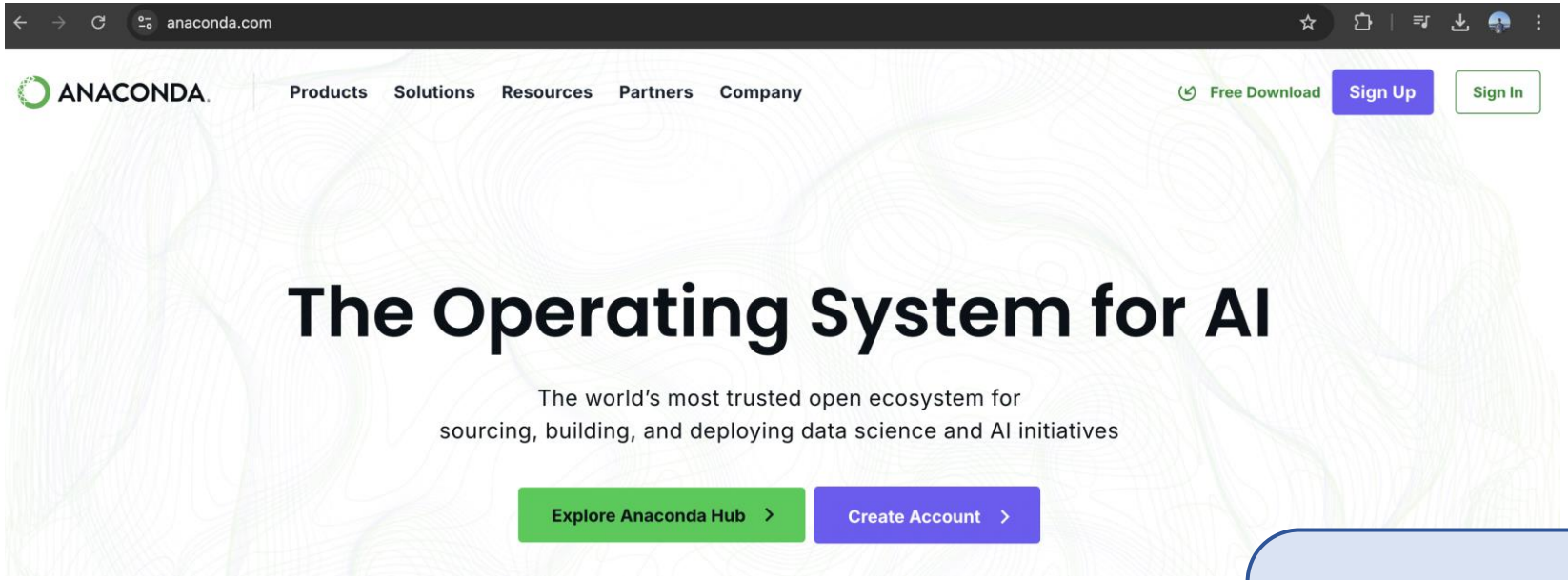
❖ Getting Started



Ref: https://www.gradio.app

Gradio

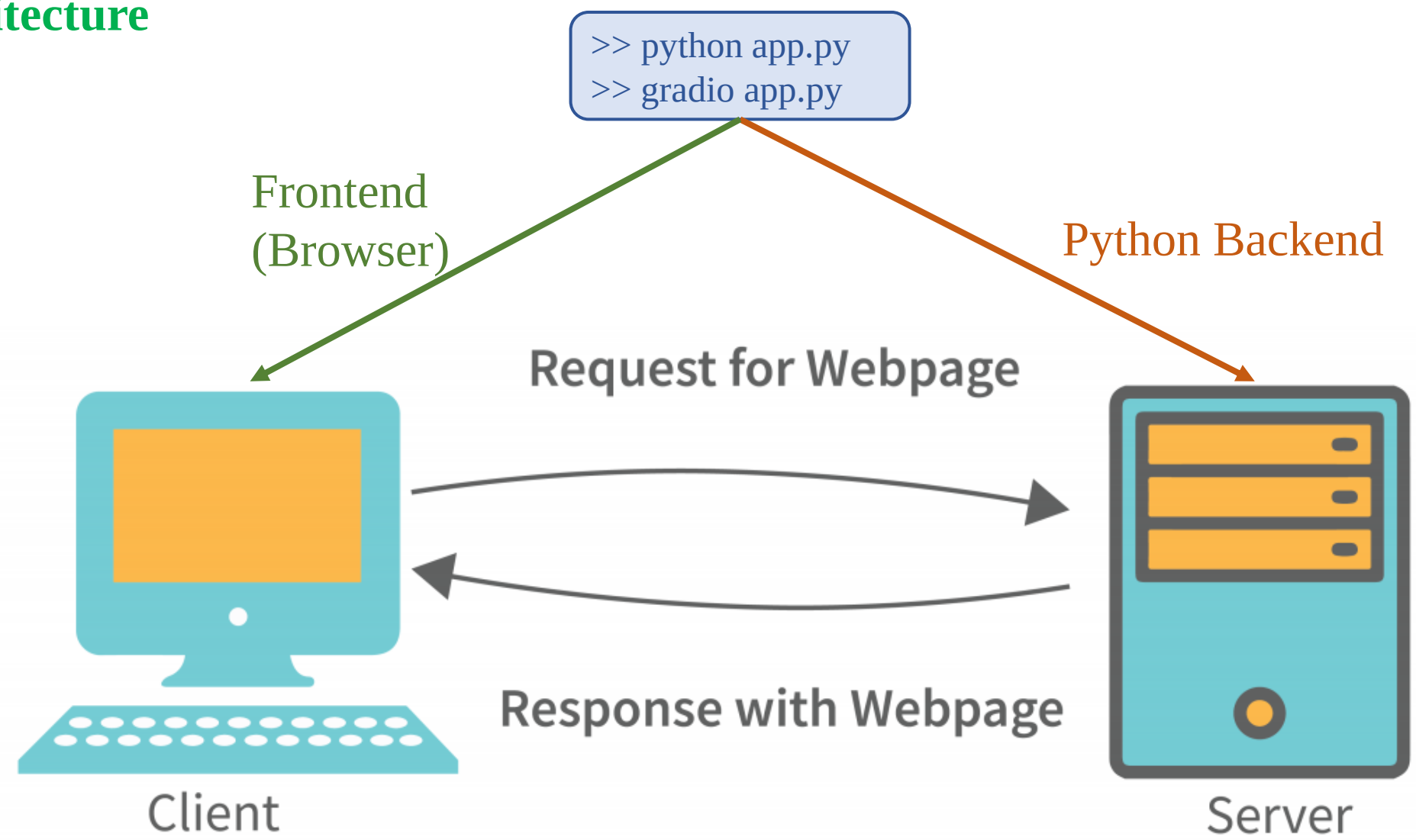
❖ Installation




```
>> conda create –n gradio_env python=3.11.9
>> conda activate gradio _env
>> pip3 install streamlit==4.44.1
```

Gradio

❖ Architecture



Gradio

❖ First app

```python
import gradio as gr

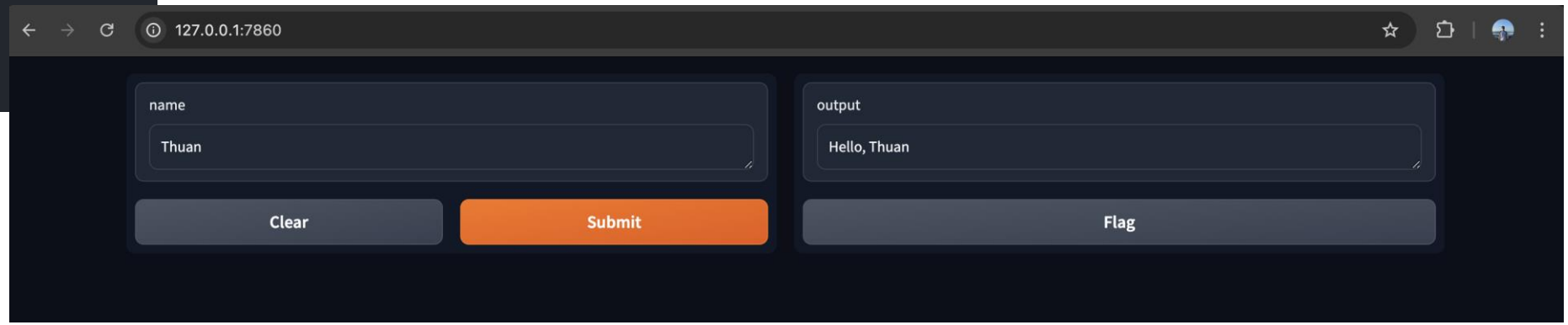
def greet(name):
    return "Hello, " + name

demo = gr.Interface(
    fn=greet,
    inputs=["text"],
    outputs=["text"],
)

demo.launch()
```

```
PROBLEMS 6  OUTPUT  TERMINAL  PORTS  JUPYTER  DEBUG CONSOLE
(aio-mlops-w1) thuannd@MacNaN gradio % python app.py
Running on local URL:  http://127.0.0.1:7860

To create a public link, set `share=True` in `launch()`.
```



Gradio with Interface

Interface

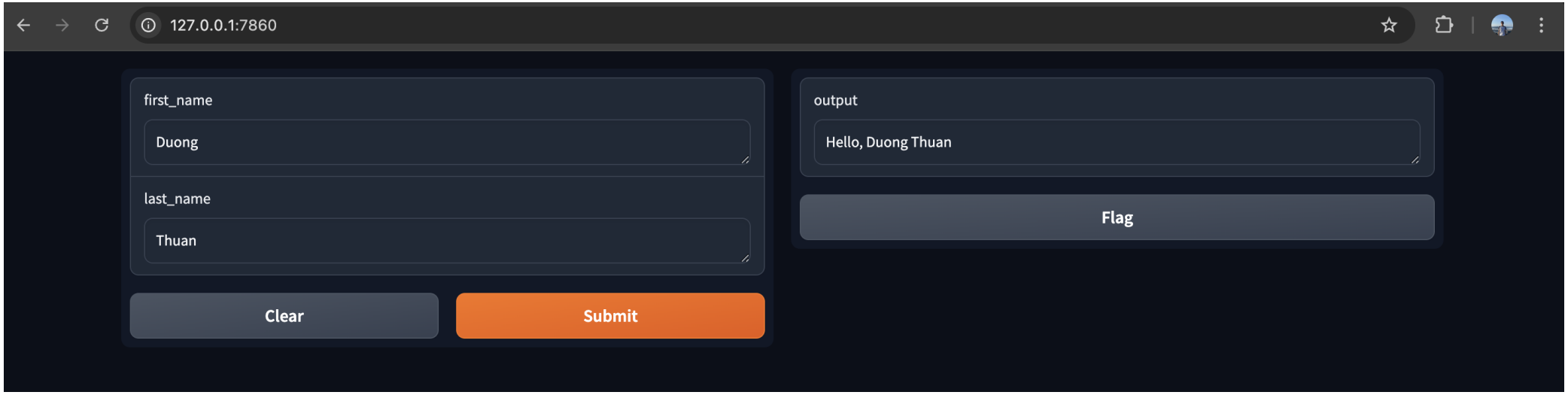
❖ Input element

```python
import gradio as gr

def greet(first_name, last_name):
    return "Hello, " + first_name + " " + last_name

demo = gr.Interface(
    fn=greet,
    inputs=["text", "text"],
    outputs=["text"],
)

demo.launch()
```



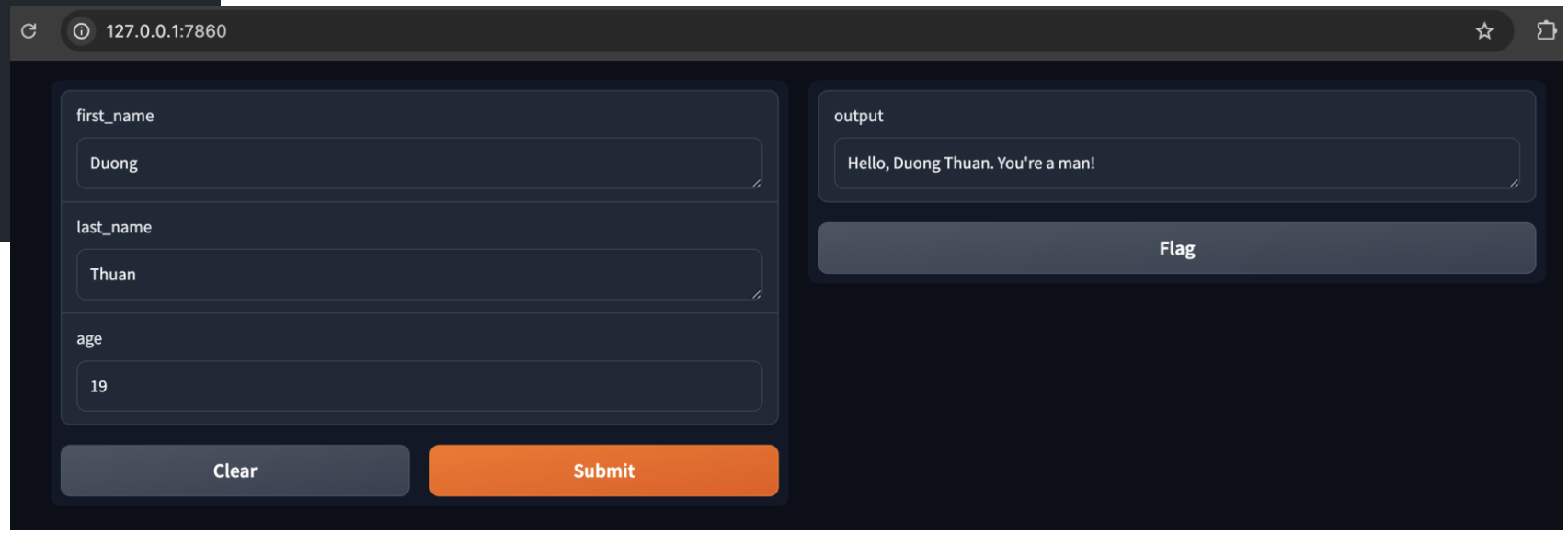
Interface

❖ Input element

```python
import gradio as gr

def greet(first_name, last_name, age):
    getting = "Hello, " + first_name + " " + last_name
    if age < 18:
        getting += ". You're a young!"
    else:
        getting += ". You're a man!"
    return getting

demo = gr.Interface(
    fn=greet,
    inputs=["text", "text", "number"],
    outputs=["text"],
)
demo.launch()
```



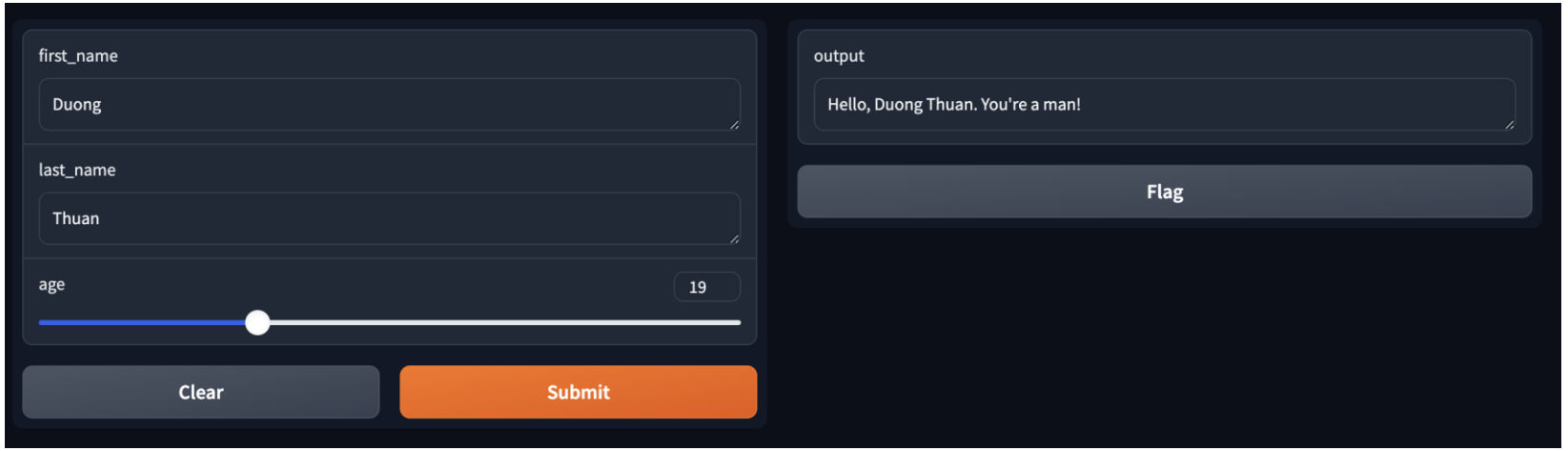
Interface

❖ Input element

```python
import gradio as gr

def greet(first_name, last_name, age):
    getting = "Hello, " + first_name + " " + last_name
    if age < 18:
        getting += ". You're a young!"
    else:
        getting += ". You're a man!"
    return getting

demo = gr.Interface(
    fn=greet,
    inputs=["text", "text", gr.Slider(1, 60, step=1)],
    outputs=["text"],
)
demo.launch()
```



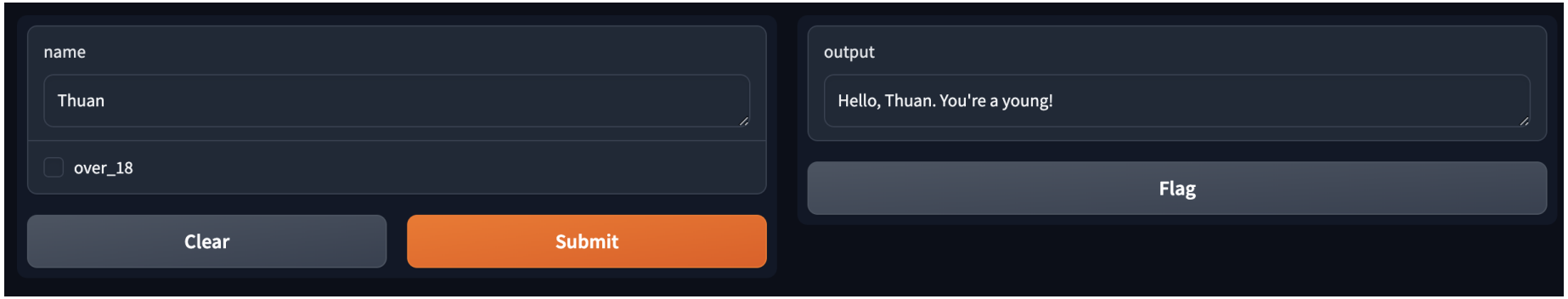
Interface

❖ Input element

```python
import gradio as gr

def greet(name, over_18, age):
    getting = "Hello, " + name
    if not over_18:
        getting += ". You're a young!"
    else:
        getting += ". You're a man!"
    return getting

demo = gr.Interface(
    fn=greet,
    inputs=["text", "checkbox"],
    outputs=["text"],
)
demo.launch()
```



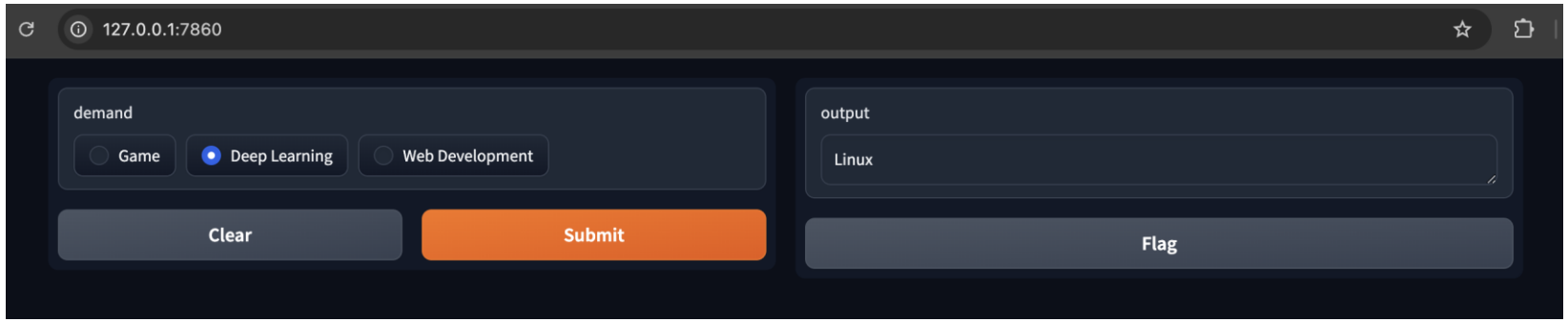
Interface

❖ Input element

```python
import gradio as gr

def choice_device(demand):
    if demand == "Game":
        return "Windows"
    elif demand == "Deep Learning":
        return "Linux"
    else:
        return "MacOS"

demo = gr.Interface(
    fn=choice_device,
    inputs=[gr.Radio(["Game", "Deep Learning", "Web Development"])],
    outputs=["text"],
)
demo.launch()
```



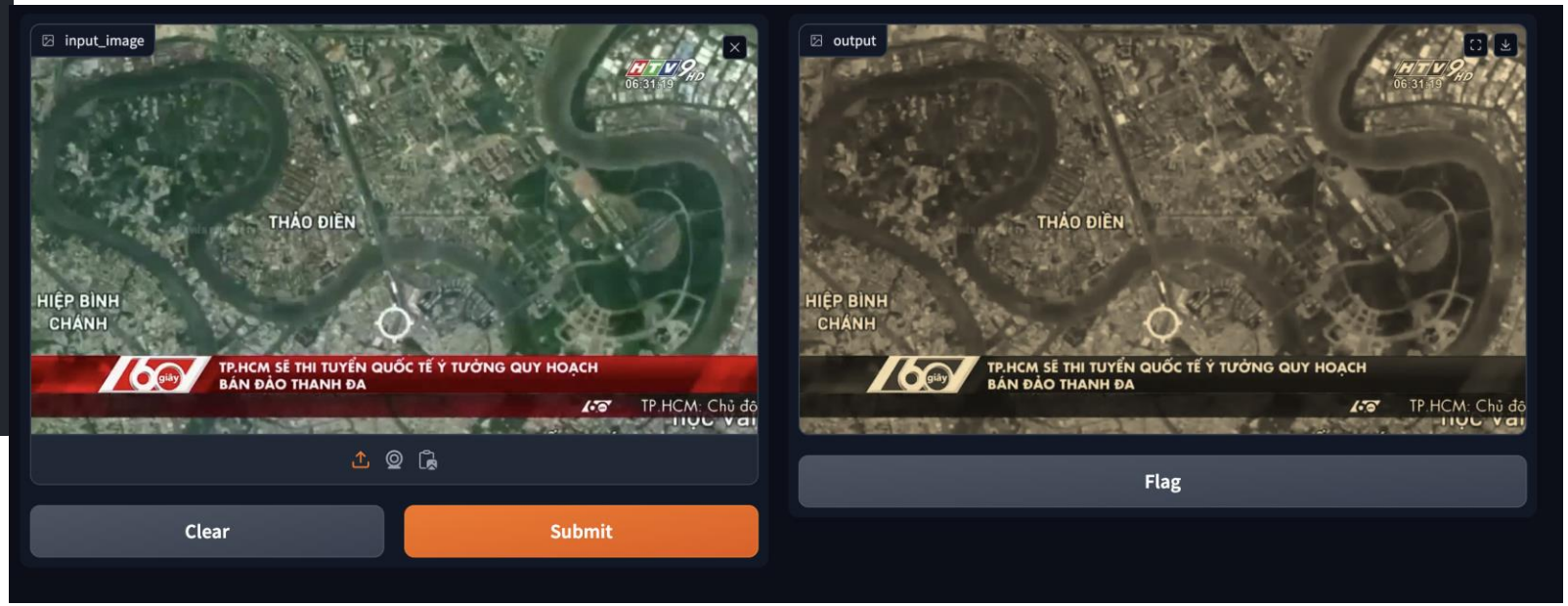
Interface

❖ Input element

```python
import numpy as np
import gradio as gr

def filter(input_image):
    sepia_filter = np.array([
        [0.393, 0.769, 0.189],
        [0.349, 0.686, 0.168],
        [0.272, 0.534, 0.131]
    ])
    sepia_img = input_image.dot(sepia_filter.T)
    sepia_img /= sepia_img.max()
    return sepia_img

demo = gr.Interface(
    fn=filter,
    inputs=["image"],
    outputs=["image"],
)
demo.launch()
```



Gradio with Block

Block

❖ Click event

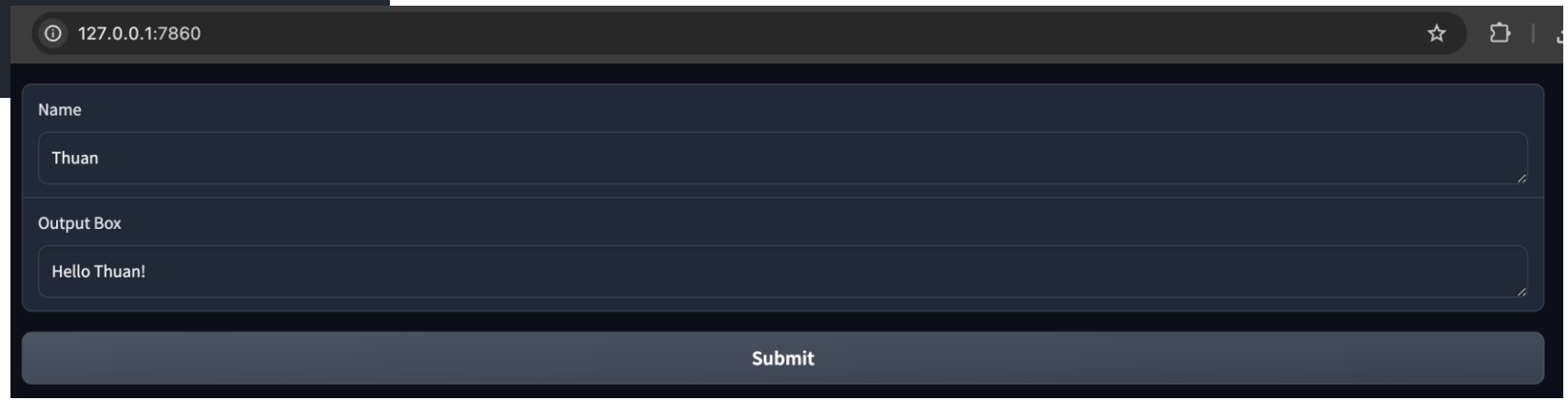
```python
import gradio as gr

def greet(name):
    return "Hello " + name + "!"

with gr.Blocks() as demo:
    input = gr.Textbox(label="Name")
    output = gr.Textbox(label="Output Box")

    greet_btn = gr.Button("Submit")
    greet_btn.click(fn=greet, inputs=input, outputs=output)

demo.launch()
```



Block

❖ Change event

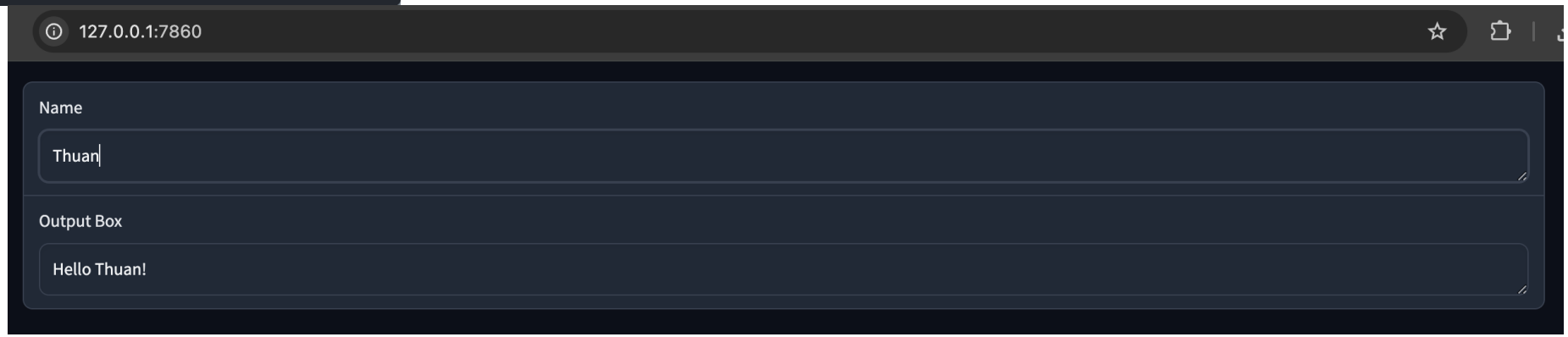
```python
import gradio as gr

def greet(name):
    return "Hello " + name + "!"

with gr.Blocks() as demo:
    input = gr.Textbox(label="Name")
    output = gr.Textbox(label="Output Box")

    input.change(fn=greet, inputs=input, outputs=output)

demo.launch()
```



Block

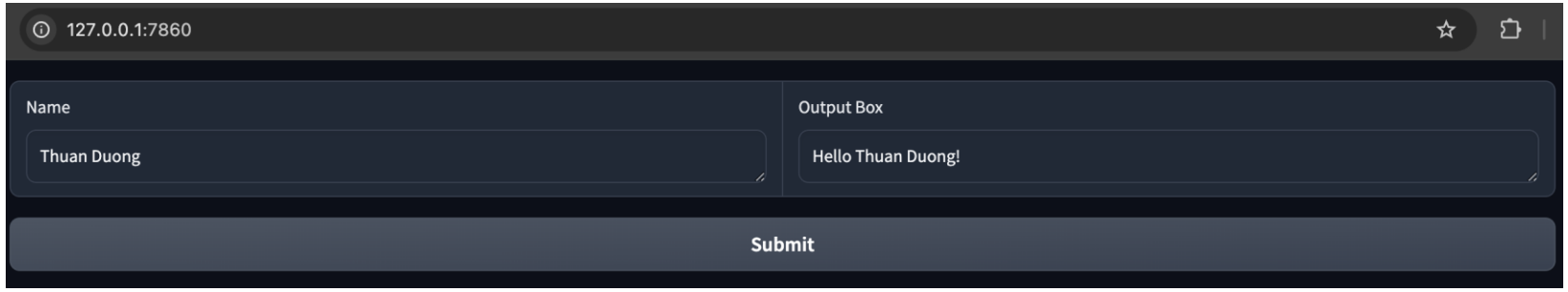
❖ Layout

```python
import gradio as gr

def greet(name):
    return "Hello " + name + "!"

with gr.Blocks() as demo:
    with gr.Row():
        input = gr.Textbox(label="Name")
        output = gr.Textbox(label="Output Box")
    greet_btn = gr.Button("Submit")
    greet_btn.click(fn=greet, inputs=input, outputs=output)

demo.launch()
```



Block

❖ Layout

```python
import gradio as gr

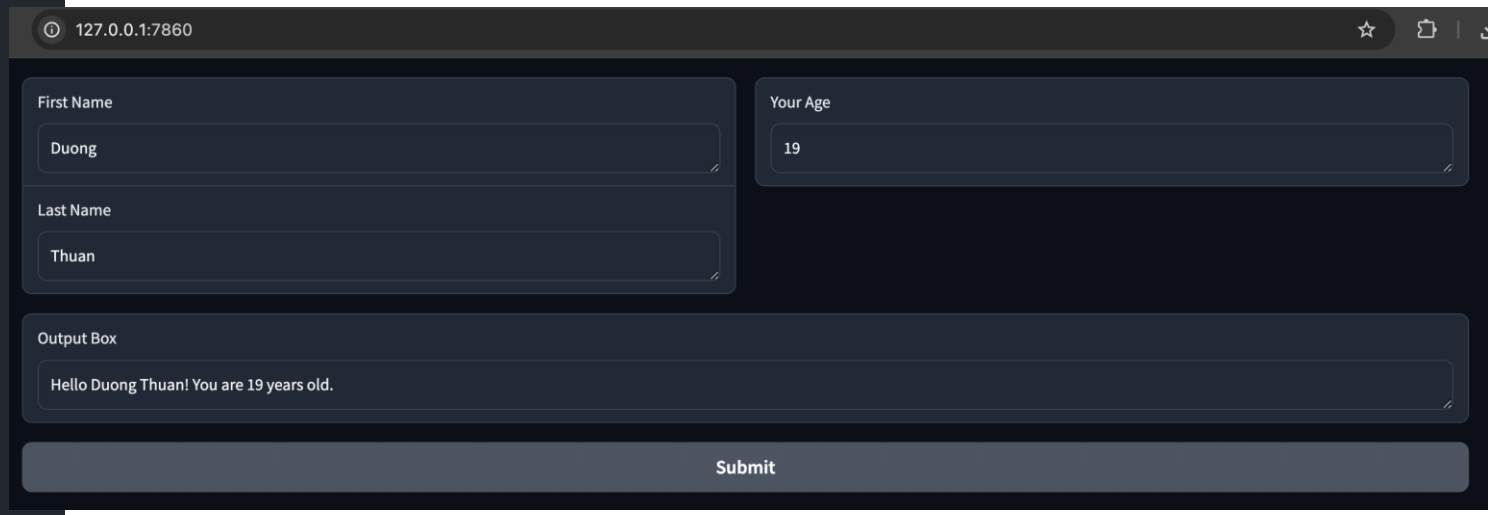
def greet(first_name, last_name, age):
    output_1 =  "Hello " + first_name + " " + last_name
    output_2 = "! You are " + age + " years old."
    return output_1 + output_2

with gr.Blocks() as demo:
    with gr.Row():
        with gr.Column():
            first_name = gr.Textbox(label="First Name")
            last_name = gr.Textbox(label="Last Name")
        with gr.Column():
            age = gr.Textbox(label="Your Age")

    with gr.Row():
        output = gr.Textbox(label="Output Box")

    greet_btn = gr.Button("Submit")
    greet_btn.click(fn=greet,
                    inputs=[first_name, last_name, age],
                    outputs=[output])

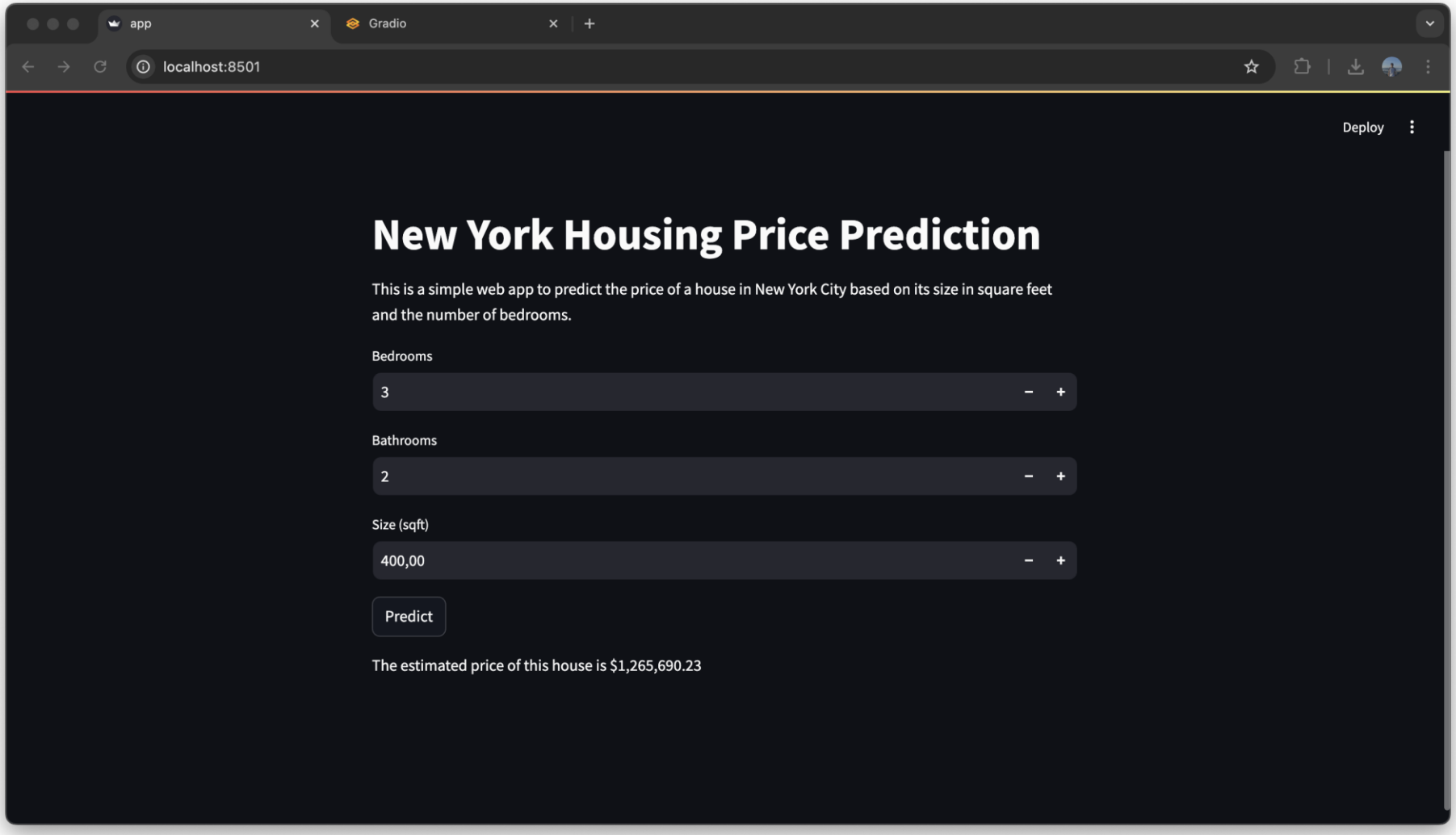
demo.launch()
```



Hands-on

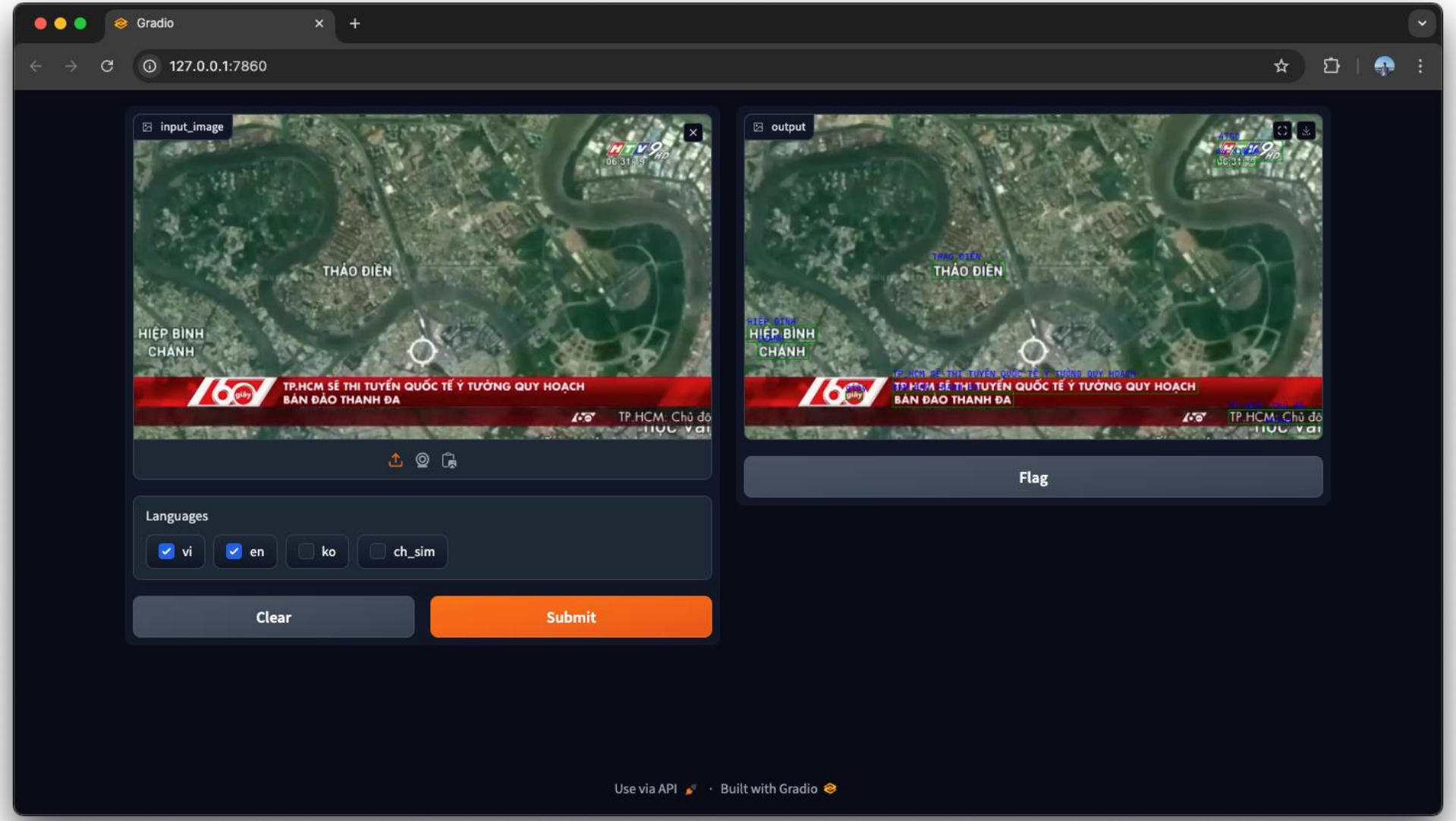
Hands-on

❖ Housing Price



Hands-on

❖ OCR

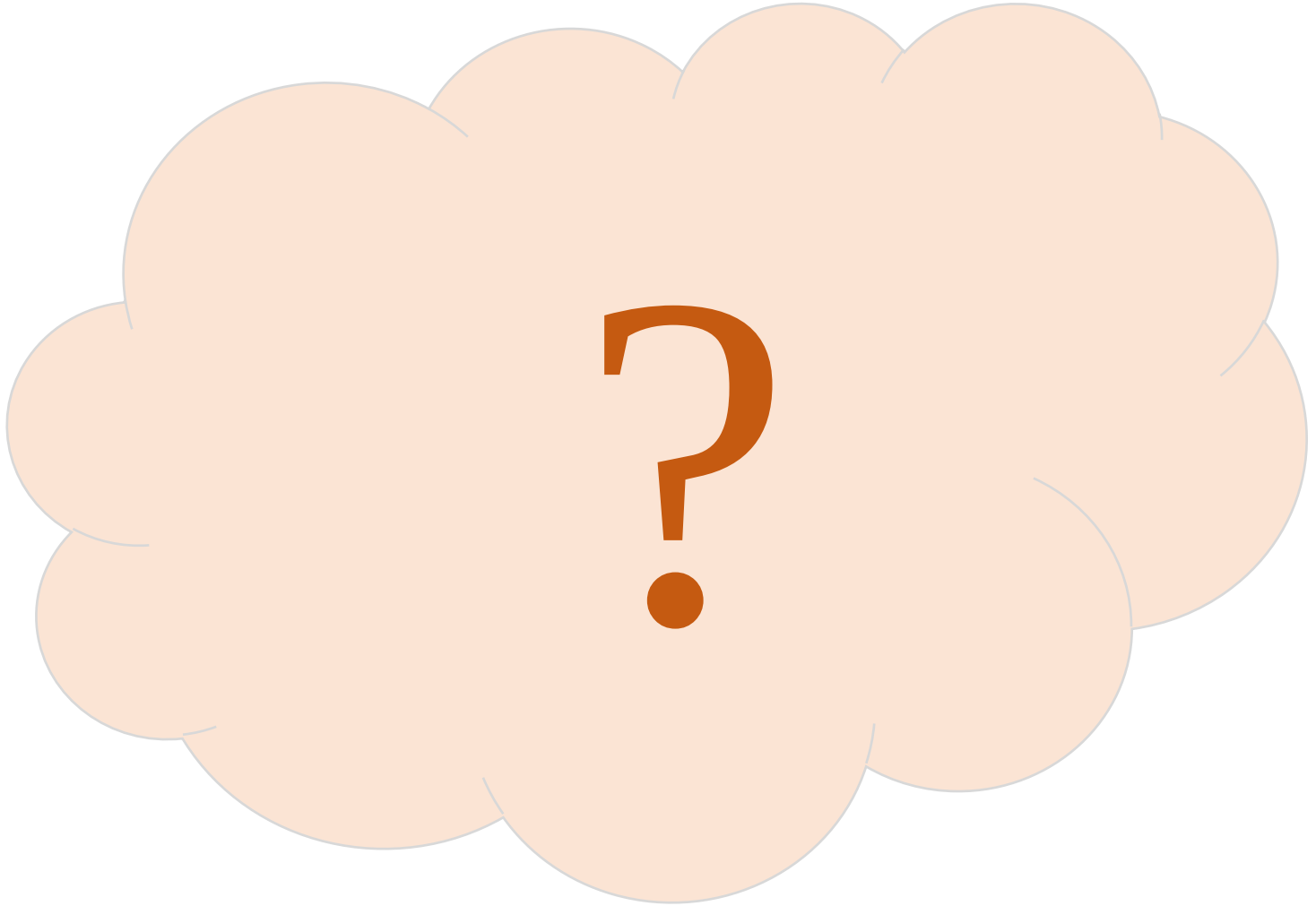

Gradio
127.0.0.1:7860
input_image
output
THẢO ĐIỀN
HIỆP BÌNH
CHÁNH
TP.HCM SẼ THI TUYỂN QUỐC TẾ Ý TƯỞNG QUY HOẠCH
BÁN ĐẢO THANH ĐA
TP.HCM. Chủ đề
Languages
vi
en
ko
ch_sim
Clear
Submit
Flag
Use via API
Built with Gradio

Conclusion

Conclusion


Machine Learning Model
Deep Learning Model
Functions
gradio
Streamlit
Application 1
Application 2
Application …

Question




Thank you!